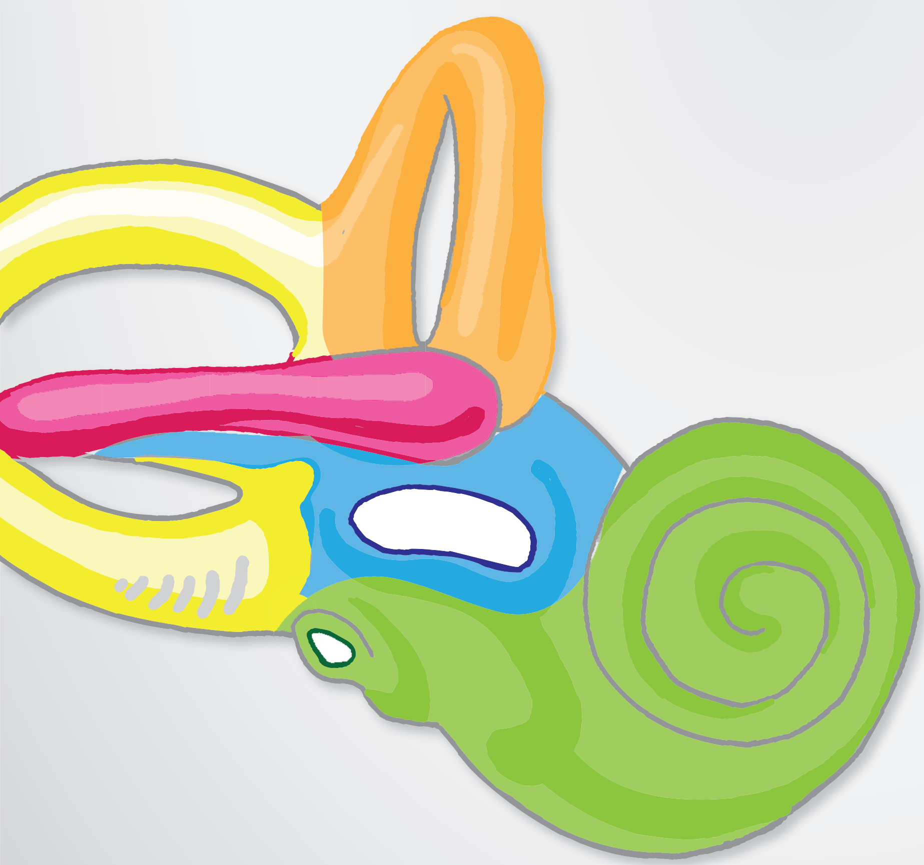
13 Senses and skin

David Kachlík
Radovan Hudák
Ondřej Volný
Adam Whitley



A **sense** is the ability of an organism to **accept and interpret signals from the the external or internal environment**. **Exteroceptors** detect signals from the **external environment**, such as the concentration of certain chemical substances in a medium, light and vibration of particles in air. Exteroceptors provide the senses of smell, taste, vision, hearing, touch and pain. **Interoceptors** monitor the internal environment of the organism. They include osmoreceptors, chemoreceptors, baroreceptors and proprioceptors and pain and tactile receptors located within internal organs, muscles, fasciae and vessels. **The perception of balance** requires signals from both exteroceptors (in the vestibular organ) and interoceptors (proprioceptors providing information on the position of the body).

Senses and their organs

1 **Vision** – eye
2 **Hearing** – outer, middle and inner ears
3 **Balance** – inner ear
4 **Taste** – tongue, palate
5 **Touch, pain, proprioception** – skin, muscles
6 **Internal environment** – organs, fasciae

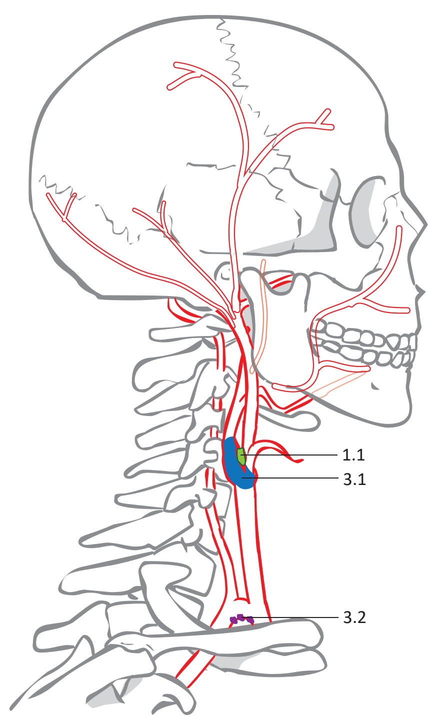
1.1 Internal environment – Interoceptors

In contrast with exteroceptors, **interoceptors monitor the internal environment of an organism**. They include: **chemoreceptors** that monitor the blood acidity (pH) and the level of blood oxygen and carbon dioxide, **osmoreceptors** that monitor the osmotic pressure of blood and **baroreceptors** that monitor blood pressure. In a broader sense, interoceptors include receptors of proprioception, which provide information on the position of the body and touch and pain receptors in the internal organs, muscles, fasciae and vessels.

1 **Chemoreceptors**
- 1.1 **Carotid body** (*glomus caroticum*)
  – is located at the carotid bifurcation
  – monitors the blood oxygen level
- 1.2 **Area postrema**
  – is located on the floor of the fourth ventricle
  – monitors blood pH

2 **Osmoreceptors**
- 2.1 **Vascular organ of lamina terminalis**
  (*organum vasculosum laminae terminalis*)
  – is located on the rostral wall of the third ventricle
  – monitors the osmolality of plasma
- 2.2 **Supraoptic and paraventricular nuclei of hypothalamus**
  (*nucleus supraopticus et paraventricularis hypothalami*)
  – are located inside the hypothalamus
  – monitor the osmolality of plasma

3 **Baroreceptors**
- 3.1 **Carotid sinus** (*sinus caroticus*)
  – is located at the beginning of the internal carotid artery
- 3.2 **Paraaortic bodies**
  (*corpora aortica / glomera aortica*)
  – are located within the aortic arch
- 3.3 **Atrial baroreceptors / atrial volume receptors**
  – are located within the wall of the right atrium



The olfactory part of the nasal mucosa occupies an area of approximately 5cm².

The vomeronasal organ / Jacobson's organ *(organum vomeronasale)* is a rudimentary organ on the inferior and anterior margin of the nasal septum. It is a chemosensory organ for pheromones.

Around 5000 taste buds are present on the mucosa of the tongue. The majority of them are located within the vallate papillae (250 on each vallate papilla) and the foliate papillae. A few are situated in the fungiform papillae.

Umami is a Japanese term meaning palatable or delicious. The taste of unami is mediated by natrium glutamate.

The supraoptic and paraventricular nuclei of the hypothalamus contain osmoreceptors. Their stimulation evokes the feeling of thirst and the secretion of antidiuretic hormone.

Olfaction is closely related to taste perception. Receptors of these senses are rather similar. These senses are fused in lower animal species.

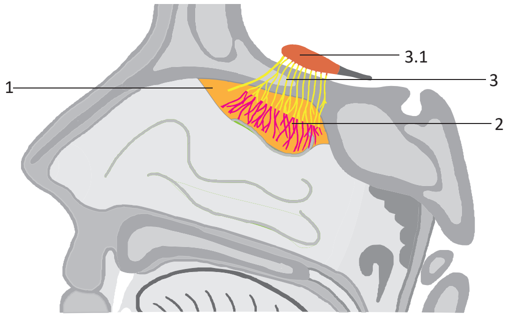
Clinical notes

Objective olfactometry is an examination in which individual smells are tested with one nostril and both eyes closed. A reduced perception of olfaction is termed hyposmia and an elevated perception of smell is called hyperosmia. An abnormal sensation of smell is termed parosmia. An inappropriate smell of faeces is called cacosmia.

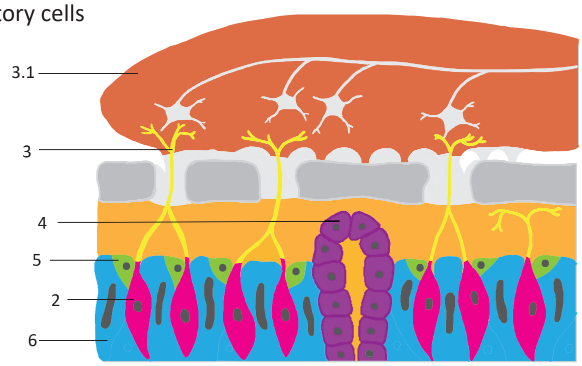
The olfactory part of the nasal mucosa (*regio olfactoria*) is located on the **roof of the nasal cavity, the nasal septum and the superior nasal concha**. Olfactory mucosa is composed of **olfactory cells and glands**. The sense of olfaction (smell) is perceived by the detection of chemicals dissolved in air or fluid (usually at low concentrations).

Structures

- 1 **Olfactory part of nasal mucosa** – specialised olfactory epithelium
- 2 **Olfactory cells** – bipolar neurons
- 3 **Olfactory nerves** (*fila olfactoria*) – bundles of axons from the olfactory cells passing through the cribriform plate (*lamina cribrosa ossis ethmoidalis*) to the olfactory bulb
  - 3.1 **Olfactory bulb** (*bulbus olfactorius*) – location of synapses between the olfactory nerves with the 2nd order neurons of the olfactory pathway
- 4 **Olfactory glands / glands of Bowman** (*glandulae olfactoriae*) – produce mucus that covers the olfactory mucosa
  – are able to concentrate and dissolve the chemicals that are perceived in the sense of olfaction
- 5 **Basal epithelial cells** – stem cells which differentiate into olfactory cells
- 6 **Supporting epithelial cells** – tall cells located between the olfactory cells



Sagittal section of the nasal cavity showing the olfactory part of the nasal mucosa



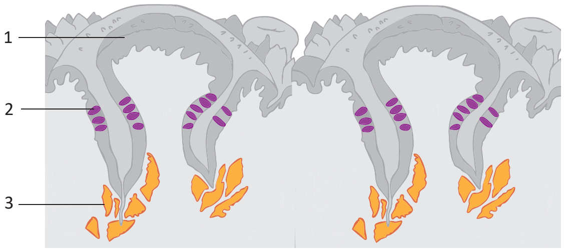
The olfactory mucosa and nerve in detail

1.3 **Gustatory organ** – Organum gustatorium

Taste buds are the **receptor cells** of the gustatory organ. The majority of the taste buds are situated in the **mucosa of** the **lingual vallate papillae**. However, they are also found in the mucosa of the soft palate, posterior wall of the pharynx, glosso-epiglottic folds and epiglottis. Free nerve endings can also operate as gustatory receptors. **Five principal types of taste** are distinguished: sweet, salty, acid, bitter and umami. Each taste bud is able to perceive each type of taste individually and in combinations.

Structures

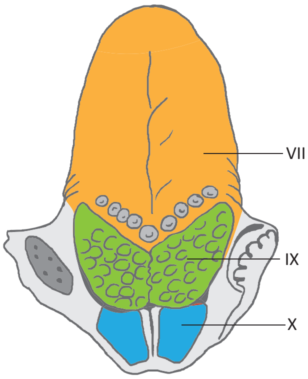
- 1 **Vallate papilla** (*papilla vallata*)
  - 1.1 **Vallum of papilla** (*vallum papillae*)
    – the external wall (similar to a rampart)
  - 1.2 **Groove of papilla** (sulcus papillae) – its walls contain the taste buds and orifices of serous salivary glands opening to the base
- 2 **Taste bud** (*gemma gustatoria / caliculus gustatorius*)
  - 2.1 **Taste pore** (*porus gustatorius*) – a superficial pit over the taste bud
- 3 **Gustatory glands / von Ebner's glands** (*glandulae gustatoriae*)
  – small serous salivary glands
  – produce saliva that dissolves the chemicals of taste and rinses the taste buds



A section through the vallate papillae

Innervation

- **VII Chorda tympani** – provides taste for the dorsum of the tongue
  – a branch from the facial nerve
  – its impulses travel partially together with the lingual nerve to the tongue
- **IX Glossopharyngeal nerve** – povides taste for the root of the tongue and vallate papillae
- **X Vagus nerve** – provides taste for the epiglottic valleculae



Sensory innervation of the tongue

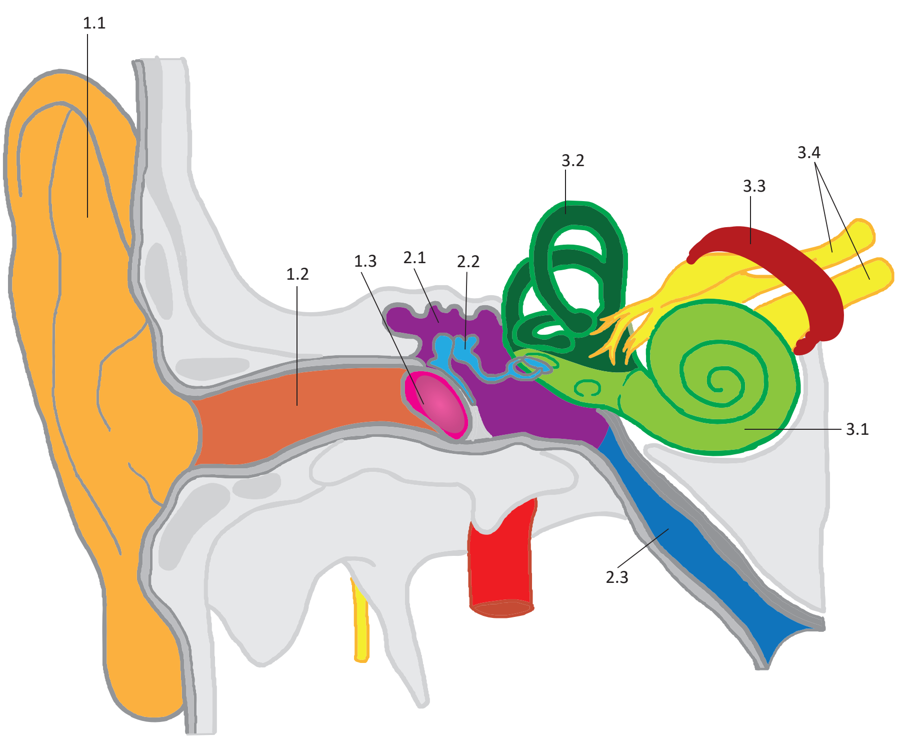
The **ear** (**auris**) is a complex sensory organ. It is divided into three parts: the external ear, middle ear and internal ear. The **external ear** consists of the auricle, the external acoustic meatus, and the tympanic membrane. The **middle ear** consists of the tympanic cavity, which houses the tympanic membrane and auditory ossicles. The **internal ear** consists of the bony and membranous labyrinths. All three parts of the ear are involved in the perception of sound. The perception of balance, position and angular movements of the head are all functions of the vestibulum in the inner ear.

Principal parts of the ear

1 **External ear** (*auris externa*)
- 1.1 **Auricle/pinna** (*auricula*)
- 1.2 **External acoustic meatus** (*meatus acusticus externus*)
- 1.3 **Tympanic membrane** (*membrana tympanica*)

2 **Middle ear** (*auris media*)
- 2.1 **Tympanic cavity** (*cavitas tympani*)
- 2.2 **Auditory ossicles** (*ossicula auditus*)
- 2.3 **Pharyngotympanic tube / auditory tube / Eustachian tube** (*tuba auditiva*)

3 **Internal ear** (*auris interna*) – consists of the bony and membranous labyrinth
- 3.1 **Cochlear part** – consists of the bony and membranous cochlea
- 3.2 **Vestibular part** – consists of the vestibule containing the utricle and saccule and three semicircular canals containing the three semicircular ducts
- 3.3 **Internal acoustic meatus** (*meatus acusticus internus*)
- 3.4 **Vestibulocochlear nerve** (*nervus vestibulocochlearis, n. VIII*)



Frontal section of the principal parts of the ear

The ear is called **us (otos)** in Greek.

The intrinsic muscles of the external ear are rudimentary in humans. This group of muscles includes the tragicus, antitragicus, helicis major, helicis minor, oblique muscle of the auricle *(m. obliquus auriculae)*, transverse muscle of the auricle *(m. transverses auriculae*) and the pyramidal muscle of the auricle *(m. pyramidalis auriculae)*. All are innervated by the facial nerve.

Hairs of the tragus *(tragi)* are hairs growing in the external acoustic meatus after the 30th year of age.

Dimensions and positions of the tympanic membrane (approximately)
– **size**: 9x10 mm
– **thickness**: 0,1 mm
– **surface**: 55 mm^2
– **declination**:
50° in the sagittal plane
– **inclination**:
45° in the transverse plane

The clinical synonyms for the light reflex in the otoscopic examination are: cone of light, light reflex, Politzer's luminous cone, red reflex and Wilde's triangle.

The flaccid part / Shrapnell membrane *(pars flaccida)* is a part of the tympanic membrane, 5 mm^2 large, located between the anterior and posterior mallear folds.

Clinical notes

Othematoma is a haematoma of the auricle located between the skin and elastic cartilage of the auricle accompanied with oedema.

The dorsocaudal part of the external acoustic meatus is innervated by the vagus nerve (auricular branch). This area is called the zone of Ramsay-Hunt. In herpes zoster infection, blisters can appear in this zone (herpes zoster oticus).

Bezold's triad of the tympanic membrane:
– mallear prominence
– mallear stria
– light reflex

Otitis media / mesotitis is term that refers to inflammation of the middle ear. Its treatment can involve myringotomy, which is an incision of the tympanic membrane (a form of paracentesis). The incision is made in the lower posterior quadrant as no anatomical structures are located in this region and so the risks of complications are minimal.

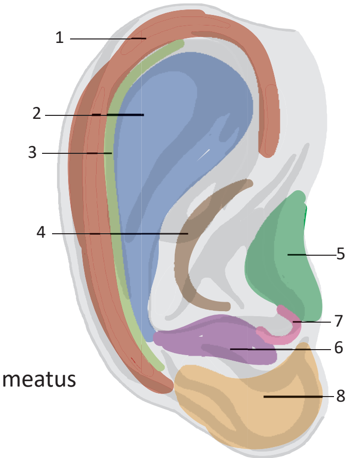
External ear

Auricle/pinna (*auricula*)

– is composed of elastic cartilage and skin adhering to the underlying perichodrium (firmly on the ventral surface and loosely on the dorsal surface)

Parts:

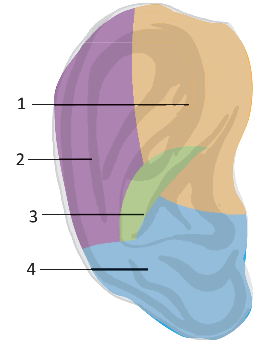
- 1 **Helix** – an elevated arched margin of the auricle located on the dorsal and cranial aspect
- 2 **Antihelix** – an elevated eminence of cartilage, located ventral to the helix,
  – bifurcates ventrocranially into the the two crura (limbs) of the antihelix (*crura antihelicis*)
- 3 **Scapha** – a narrow groove between the helix and antihelix on the anterior surface of the auricle
- 4 **Concha of auricle** (*concha auriculae*) – a depression on the anterior surface of the auricle
  – its upper part (cymba conchae) is located between the helix and antihelix
  – its inferior part (cavity of concha) serves as the entrance of the external acoustic meatus
- 5 **Tragus** – a process in front of the concha that partially covers the entrance of the external acoustic meatus
- 6 **Antitragus** – a process below the concha facing the tragus
  – partially covers the entrance of the external acoustic meatus
- 7 **Intertragic notch** (*incisura intertragica*) – a notch between the tragus and antitragus
- 8 **Lobule of auricle** (*lobulus auriculae*) – the earlobe, does not contain cartilage



Parts of the auricle

Ligaments: anterior, superior and posterior ligaments of the auricle (*lig. auriculare anterius, superius, posterius*)
Muscles: auricularis anterior, superior and posterior and temporoparietalis
Motor innervation: facial nerve
Sensory innervation:

- 1 **Superior anterior two thirds** – anterior auricular branches of auriculotemporal nerve
- 2 **Superior posterior** – lesser occipital nerve
- 3 **Concha of auricle** – auricular branch of vagus nerve
- 4 **Inferior** – great auricular nerve



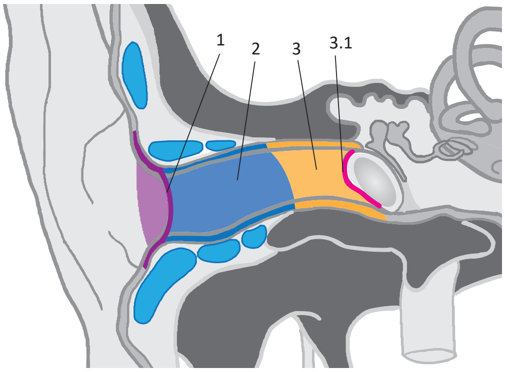
Somatosensory innervation of the auricle

External acoustic meatus (*meatus acusticus externus*)

– has a waved shape, runs ventromedially, then medially and then again ventromedially
– both meati are directed ventrally (at an angle of 160°) and caudally (cranially convex)
– has a length of approximately 22 mm
– the anterior wall is in close proximity to the parotid gland and temporomandibular joint

Parts:

- 1 **External acoustic pore / aperture** (*porus acusticus externus*)
- 2 **Cartilaginous external acoustic meatus** (*meatus acusticus externus cartilaginous*)
  – the external cartilaginous 2/3 of the meatus
  – hyaline cartilage opens ventrally with the tragal lamina (*lamina tragi*) and caudally through the notches of Santorini (*incisurae*)
  – the skin of the meatus is firmly attached to the perichondrium
  – the most narrow spot is at the transition of the bony part to the cartilaginous part (the *isthmus*), foreign bodies can get stuck here
- 3 **Bony external acoustic meatus** (*meatus acusticus externus osseus*)
  – the internal bony 1/3 of the meatus
  - 3.1 **Tympanic sulcus and notch** (*sulcus tympanicus et incisura tympanica*)
    – a circular groove with a cranial notch
- 4 **Ceruminous glands** (*glandulae ceruminosae*) – apocrine glands
  – produce cerumen, which has a protective function



Frontal section of the external acoustic meatus

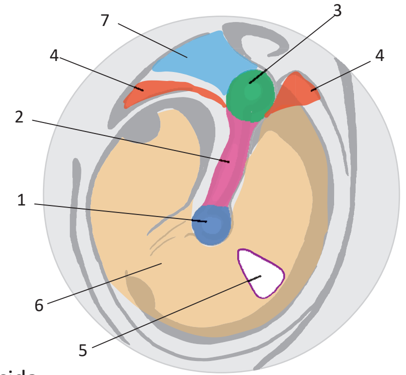
Sensory innervation:
1 **Auricular branch of vagus nerve**
2 **Nerve to external acoustic meatus** (*n. meatus acustici externi*) from the auriculotemporal nerve

Tympanic membrane / eardrum (*membrana tympanica / myrinx*)

– a thin fibrous plate inserted in the tympanic sulcus and notch
– has a thickened margin called the fibrocartilaginous ring (*anulus fibrocartilagineus*)

Otoscopic view – a view of the tympanic membrane from the external acoustic meatus

- 1 **Umbo of tympanic membrane** (*umbo membranae tympanicae*)
  – a convex centre elevated by the tip of the handle of the malleus
- 2 **Mallear stria** (*stria mallearis*) – a fibrous band elevated by the handle of the malleus
- 3 **Mallear prominence** (prominetia mallearis) – is located cranially on the tympanic membrane
  – an elevation formed by the lateral process of the malleus
- 4 **Anterior and posterior mallear fold** (*plica mallearis anterior et posterior*)
  – folds of mucosa stretched between the mallear prominence and encompassing the pars flaccida
- 5 **Light reflex** – an area of reflected light on the inferior posterior quadrant of the tympanic membrane
- 6 **Pars tensa** – a large, tended part of the tympanic membrane
- 7 **Pars flaccida of Shrapnell** – an upper, relaxed small part of the tympanic membrane, located cranially between the mallear folds



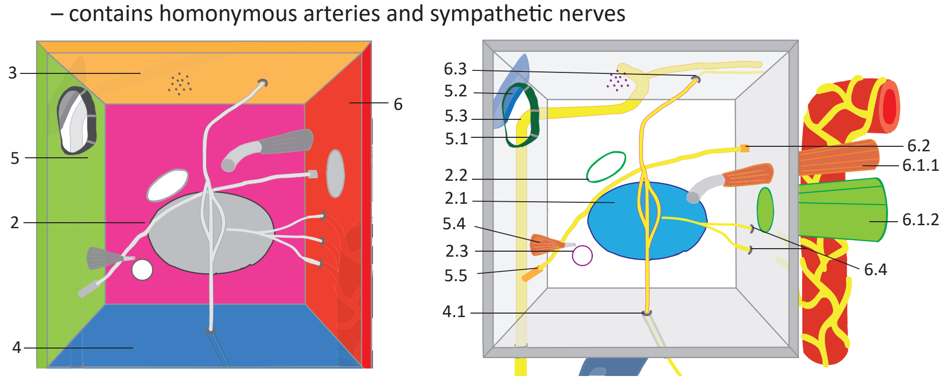
Otoscopic view of the right tympanic membrane

Middle ear (*auris media*)

Tympanic cavity (*cavitas tympani*)

Borders and structures:

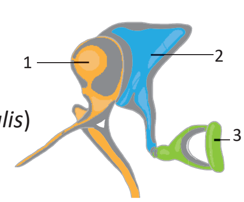
1 **Lateral wall / membranous wall** (*paries membranaceus*): the tympanic membrane
- 2 **Medial wall / labyrinthine wall** (*paries labyrinthicus*):
  - 2.1 **Promontory** (*promontorium*) – an elevation formed by the cochlea of the inner ear
  - 2.2 **Oval window** (*fenestra vestibuli*) – connects to the base of the stapes
    – the attachement of the anular ligament of the stapes (*ligamentum anulare stapediale*)
  - 2.3 **Round window** (*fenestra cochleae*) – is closed by the secondary tympanic membrane
- 3 **Roof / tegmental wall** (*paries tegmentalis*)
- 4 **Floor / jugular wall** (*paries jugularis*)
  - 4.1 **Tympanic canaliculus** (*canaliculus tympanicus*)
    – contains the tympanic nerve and anterior tympanic vessels
- 5 **Posterior wall – mastoid wall** (*paries mastoideus*)
  - 5.1 **Aditus to mastoid antrum** (*aditus ad antrum mastoideum*)
    – the entrance to the mastoid cells
  - 5.2 **Prominence of lateral semicircular canal** (*prominentia canalis semicircularis lateralis*)
    – an elevation formed by the lateral semicircular canal of the inner ear
  - 5.3 **Prominence of facial canal** (*prominetia canalis n. facialis*)
    – an elevation formed by the second bend of the facial canal
  - 5.4 **Pyramidal eminence** (*eminentia pyramidalis*) – a conical bony prominence
    – contains the stapedius
  - 5.5 **Canaliculus for chorda tympani** (*canaliculus chordae tympani*) – a some bony canal for the chorda tympani and posterior tympanic vessels
- 6 **Anterior wall / carotid wall** (*paries caroticus*)
  - 6.1 **Musculotubal canal** (*canalis musculotubarius*) – a bony canal divided by a septum into:
    - 6.1.1 **Canal for tensor tympani** (*semicanalis m. tensoris tympani*) – cranial half canal, tensor tympani inserts onto the malleus handle of the malleus
    - 6.1.2 **Canal for auditory tube** (*semicanalis tubae auditivae*) – caudal half canal, an air communication between the tympanic cavity and nasopharynx
  - 6.2 **Petrotympanic fissure** (*fissura petrotympanica*) – contains the chorda tympani, anterior mallear ligament of the malleus and anterior tympanic vessels
  - 6.3 **Canal for lesser petrosal nerve** (*canalis nervi petrosi minoris*)
    – contains the lesser petrosal nerve (n. IX) and superior tympanic vessels
  - 6.4 **Caroticotympanic canaliculi** (*canaliculi caroticotympanici*)
    – contains homonymous arteries and sympathetic nerves

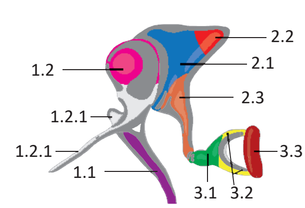


Scheme of the tympanic cavity: lateral view after removal of the tympanic membrane

Auditory ossicles (*ossicula auditus*)

- 1 **Malleus** – the hammer
  - 1.1 **Handle of malleus** (*manubrium mallei*)
  - 1.2 **Head of malleus** (*caput mallei*)
    - 1.2.1 **Anterior and lateral processes** (*processus anterior et lateralis*)
  - 1.3 **Neck of malleus** (*collum mallei*)
- 2 **Incus** – the anvil
  - 2.1 **Body of incus** (*corpus incudis*)
  - 2.2 **Short limb** (*crus breve*)
  - 2.3 **Long limb** (*crus longum*)
  - 2.4 **Lenticular process** (*processus lenticularis*)
- 3 **Stapes** – the stirrup
  - 3.1 **Head of stapes** (*caput stapedis*)
  - 3.2 **Anterior and posterior limbs** (*crus anterius et posterius*)
  - 3.3 **Base of stapes, Footplate** (*basis stapedis*)





The auditory tube / Eustachian tube has two parts: a lateral bony part and a medial cartilaginous part. Its function is to equalise the air pressure between the pharynx and tympanic cavity. The auditory tube is more horizontal, shorter and wider in children.

The tympanic cavity has a sand-glass shape extending in the epitympanium and caudally as the hypotympanic recess.

The epitympanic recess / attic *(epitympanum / epitympanon / atticus)* is the cranial extension of the tympanic cavity above the upper margin of the tympanic membrane containing the head of the malleus and body of the incus. It can be approached from outside via a bony wall called the scutum.

Muscles of the tympanic cavity:
1 **Stapedius**: innervated by the facial nerve
2 **Tensor tympani:** innervated by the trigeminal nerve

Endolymph is produced in the stria vascularis, flows into the membranous labyrinth and drains via the endolymphatic duct inside the vestibular canaliculus to the endolymphatic sac, where it is absorbed into the venous blood.
Perilymph fills the bony labyrinth.

Arterial supply of the internal ear: labyrinthine artery (from the anterior inferior cerebellar artery).
Venous drainage of the internal ear: labyrinthine veins, vein of the vestibular aqueduct, vein of the cochlear aqueduct (into the inferior petrosal sinus).

Clinical points

Otorrhea is secretion from the ear.

Otitis media / mesotitis is the most common disease of the middle ear. It is a septic inflammation that occurs most commonly in infants and toddlers. It presents with a painful ear, headache, and even a perforated eardrum.

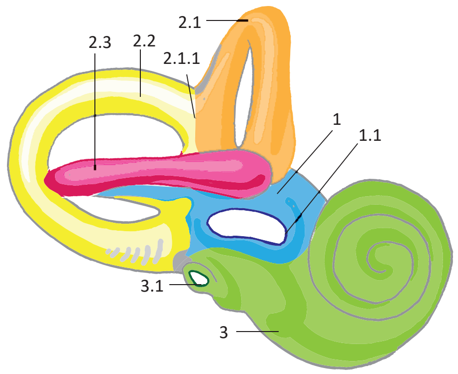
Otosclerosis is characterised by excessive ossification around the middle ear. In adulthood, ossification can involve the oval window and anular ligament of the stapes and the tympanostapedial syndesmosis can change into a synostosis. The stapes loses its ability to move within the oval window resulting in an impairment of hearing.

Hypertrophy of the pharyngeal tonsil can affect ventilation in the tympanic cavity, predisposing to secondary chronic mesotitis.

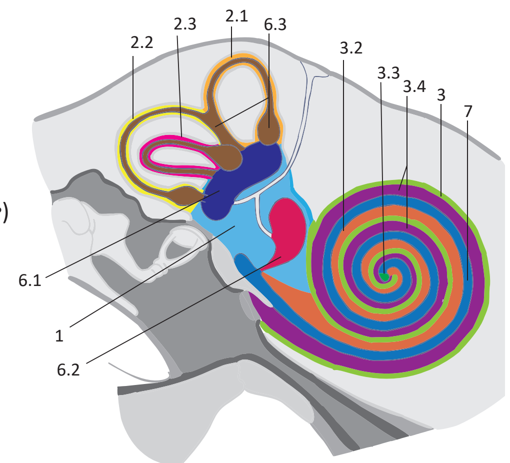
Internal ear / vestibulocochelar organ (*auris interna / organum vestibulocochleare*)

Osseous labyrinth (*labyrinthus osseus*)

- 1 **Vestibule** (*vestibulum*)
  - 1.1 **Oval window** (*fenestra vestibuli*)
  - 1.2 **Elliptical and spherical recess** (*recessus ellipticus et sphericus maculae cribrosae*)
  - 1.3 **Internal opening of vestibular canaliculus** (*apertura interna canaliculi vestibuli*)
- 2 **Semicircular canals** (*canales semicirculares*) – consist of the bony, ampulla and limb
  - 2.1 **Anterior semicircular canal** (*canalis semicircularis anterior*)
    – perpendicular to the long axis of the petrous part of the temporal bone
    - 2.1.1 **Common bony limb** (*crus osseum commune*)
      – a common limb for the anterior and posterior semicircular canals
  - 2.2 **Posterior semicircular canal** (*canalis semicircularis posterior*
    – is parallel to the long axis of the petrous part of the temporal bone
  - 2.3 **Lateral semicircular canal** (*canalis semicircularis lateralis*) – oriented 30° to the horizontal plane and contains the simplex bony limb (*crus osseum simplex*)
- 3 **Cochlea** – cupula and basis of the cochlea (2 ¾ of the cochlea threads)
  - 3.1 **Round window** (*fenestra cochleae*)
  - 3.2 **Scala vestibuli**
  - 3.3 **Helicotrema** – a narrow transition from the scala vestibuli to the scala tympani in the cochlear cupula
  - 3.4 **Scala tympani**
  - 3.5 **Osseous spiral lamina** (*lamina spiralis ossea*)
    – a bony crest projecting into the spiral canal of the cochlea
  - 3.6 **Internal opening of cochlear canaliculus** (*apertura interna canaliculi cochleae*)
  - 3.7 **Modiolus** – is located within the spiral canal (*canalis spiralis modioli*)
    – contains the cochlear ganglion and longitudinal canals
- 4 **Internal acoustic meatus** (*meatus acusticus internus*)
- 5 **Perilymphatic space** (*spatium perilymphaticum*)



Lateral view of the osseous labyrinth

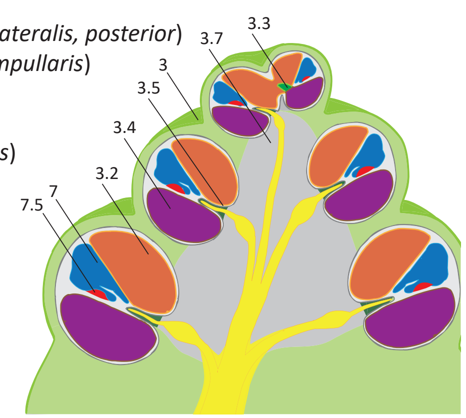


Section of the inner ear

Membranous labyrinth (*labyrinthus membranaceus*)

– surrounds the endolymphatic space (*spatium endolymphaticum*)

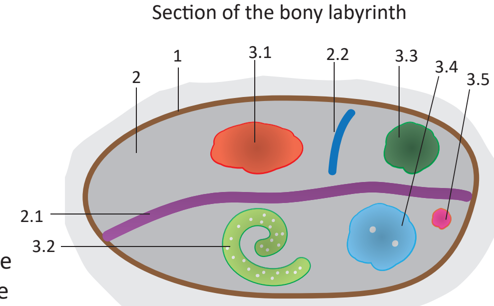
- 6 **Vestibular labyrinth** (*labyrinthus vestibularis*)
  - 6.1 **Utricle** (*utriculus*) – contains the macula of utricle, which is composed of the otolithic membrane (*membranae statoconiorum*) resting on top of special sensory cells
    – otolithic membranes contains otoliths (*otoconia*), which are crystals of $CaCO_3$ is responsible for the perception of linear horizontal movements
  - 6.2 **Saccule** (*sacculus*) – contains the macula of the saccule
    – is responsible for the perception of linear vertical movements
  - 6.3 **Anterior, lateral and posterior semicircular ducts** (*ductus semicircularis anterior, lateralis, posterior*)
    – contain ampullar crests (*cristae ampullares*) with the ampullar cupula (*cupula ampullaris*)
    – are responsible for perception of acceleration, deceleration and rotational movements of the head
- 7 **Cochlear labyrinth** (*labyrinthus cochlearis*) – scala media, cochlear duct (*ductus cochlearis*)
  - 7.1 **Vestibular surface / Reissner's membrane** (*paries / membrana vestibularis*)
  - 7.2 **External surface** (*paries externus*) – stria vascularis, spiral prominence, vas prominens (production of endolymph)
  - 7.3 **Tympanic surface, spiral membrane** (*paries tympanicus, membrana spiralis*)
    – its fibrous part is the basilar membrane (*lamina basilaris*)
  - 7.4 **Tectorial membrane** (*membrana tectoria*) – gelatinous tissue surrounding the hair of sensory cells (cochlear hair cells)
  - 7.5 **Spiral organ of Corti** (*organum spirale*) – the proper auditory organ (contains sensory cochlear hair cells)



Section of the bony labyrinth

Internal acoustic meatus (*meatus acusticus internus*)

- 1 **Internal acoustic opening** (*porus acusticus internus*)
- 2 **Fundus of internal acoustic meatus** (*fundus meatus acustici interni*)
  - 2.1 **Transverse crest** (*crista transversa*)
  - 2.2 **Vertical crest** (*crista verticalis*)
- 3 **Passage for the facial and vestibulocochlear nerves and labyrinthine vessels**
  - 3.1 **Facial area** (*area n. facialis*) – facial nerve
  - 3.2 **Cochlear nerve** (*area cochlearis*) – tractus spiralis foraminosus – cochlear nerve
  - 3.3 **Superior vestibular area** (*area vestibularis superior*) – utriculo-ampullary nerve
  - 3.4 **Inferior vestibular area** (*area vestibularis inferior*) – saccular nerve
  - 3.5 **Foramen singulare** – posterior ampullary nerve

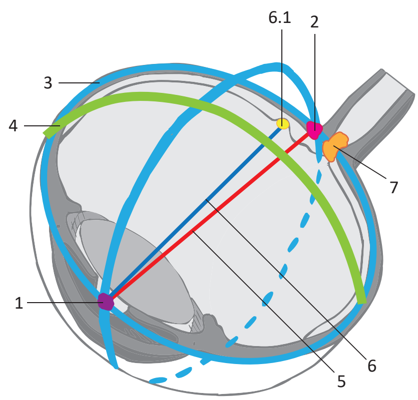


Fundus of the right internal acoustic meatus, medial view

The organ of vision is the **eyeball, which is located inside the orbit**. The eyeball is almost spherical and can be likened to two unequal hemispheres, one inside the other. The bigger hemispheres is the **sclera** (radius 12mm) and the smaller hemisphere is the **cornea** (radius 8mm). The eyeball houses the biconvex lens and contains three chambers: the **anterior and posterior chambers** filled with aqueous humour and the **postremal chamber** filled with the vitreous body. The **accessory structures of the eye** include the eyelids, eyebrows, lacrimal apparatus, conjunctiva and the extra-ocular muscles.

Eyeball (*bulbus oculi*)

- 1 **Anterior pole** (*polus anterior*)
- 2 **Posterior pole** (*polus posterior*)
- 3 **Meridians** (*meridiani*)
- 4 **Equator**
- 5 **Optic axis** (*axis opticus*) – connects the anterior and posterior poles of the eyeball
- 6 **Visual axis** (*linea visus*) – connects the centre of the viewed object to the macula
  - 6.1 **Macula** (*macula lutea*)
    - – the area of highest visual acuity
    - – located temporal to the optic axis
- 7 **Optic disc** (*discus nervi optici*)
  – the area where the optic nerve leaves the eyeball



Eyeball with orientation points and lines

Basic parts of the eyeball

Layers of the eyeball

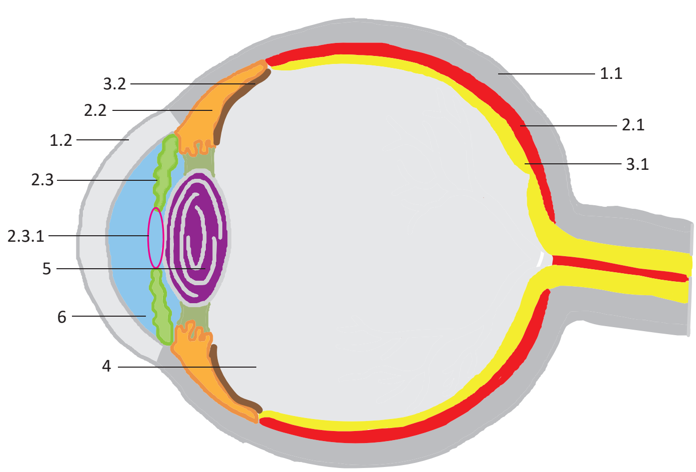
1 **Fibrous layer of eyeball** (*tunica fibrosa bulbi*) – the outer layer
- 1.1 **Sclera**
- 1.2 **Cornea**

2 **Vascular layer of eyeball** (*tunica vasculosa bulbi / uvea*) – the middle layer
- 2.1 **Choroid** (*choroidea*)
- 2.2 **Ciliary body** (*corpus ciliare*)
- 2.3 **Iris**
  - 2.3.1 **Pupil** (*papilla*)

3 **Inner layer of eyeball** (*tunica interna bulbi / retina*) – the nervous layer
- 3.1 **Optic part of retina** (*pars optica*) – contains photoreceptors
- 3.2 **Nonvisual retina** (*pars caeca*) – a blind part of retina, not containing any photoreceptors

Contents of the eyeball

- 4 **Vitreous body** (*corpus vitreum*)
- 5 **Lens**
- 6 **Chambers of eyeball** (*camerae bulbi*)



Sagittal section of the eyeball

The eyeball is approximately 25 mm long in its ventrodorsal dimension and 23.5 mm high in its craniocaudal dimension.

Ophthalmos is the Greek term for eye.
Keras (keratos) is the Greek term for cornea.
Koré is the Greek term for pupil.
Fakos is the Greek term for lens.

The pupil is analogous in function to the lens hood of a camera.

Isocoria means symmetrical pupils.
Anisocoria means asymmetrical pupils (one is wider than the other).

Bruch's membrane *(complexus basalis)* is composed of five layers. From outside to inside, the layers are: the basal lamina of the choriocapillary layer, an outer layer of collagen, a layer of elastic fibres, an inner layer of collagen and the basal lamina of the pigmented epithelium. By means of passive traction on the ciliary body, Bruch's membrane contributes to the flattening of the lens.

The orbit see page 532.

Clinical notes

Keratoplasy usually refers to corneal transplantation. The cornea can be transplanted between two unrelated individuals and is one of the most successful transplantation procedures.

The pupillary light reflex is one of the basic reflexes examined in neurology and acute care.

The direct response is a constriction of the pupil in response to light shone in the ipsilateral eye.

The consensual response is a constriction of a pupil in response to light being shone in the contralateral eye. It is physiological.

The cornea is the most sensitive tissue of the human body. When the cornea is touched, the corneal reflex results in a contraction of the orbicularis oculi causing the eye to blink.

The corneoscleral junction *(corneal limbus / limbus corneae)* is the transition between the cornea and sclera. Hyperemia of this area may signify blood congestion inside the eyeball.

CCT refers to central corneal thickness and is approximately 555 μm at the corneal vertex *(vertex corneae)*. Towards the limbus the thickness of the cornea increases up to 1 mm.

Fibrous layer of eyeball (*tunica fibrosa bulbi*) – outer fibrous layer

Sclera

1 **Trabecular tissue** (*reticulum trabeculare*) – dense connective tissue
2 **Episcleral layer** (*lamina episcleralis*), **substantia propria, suprachoroid lamina** (*lamina fusca*)
– layers of the sclera
3 **Lamina cribrosa of sclera** – the exit for the optic nerve
4 **Scleral venous sinus / Schlemm's canal** (*sinus venosus sclerae /canalis Schlemmi*)
5 **Sulcus sclerae** – a groove located at the corneoscleral junction

Cornea

– posseses a physiological astigmatism: the cornea has a vertical diameter of 11mm and a horizontal diameter of 12mm
– has an optical power of ±43 dioptres
– somatosensory innervation is provided by the branches of the ophthalmic nerve
1 **Corneoscleral junction / corneal limbus** (*limbus corneae*)
– a shallow groove at the transition between the sclera and cornea
2 **Corneal vertex** (*vertex corneae*) – the thinnest and ventral-most part of the cornea (555 µm)
Structure
1 **Corneal epithelium** (*epithelium anterius*) – stratified squamous non-keratinising epithelium
– has an extensive regeneration ability (6-day cycle)
2 **Anterior limiting lamina of Bowman** (*lamina limitans anterior*)
3 **Substantia propria** – if injured produces scars, which cause corneal opacity
4 **Posterior limiting lamina / Descemet's membrane** (*lamina limitans posterior*)
5 **Endothelium of anterior chamber** (*epithelium posterius*) – simple flat epithelium

Fibrous layer of the eyeball

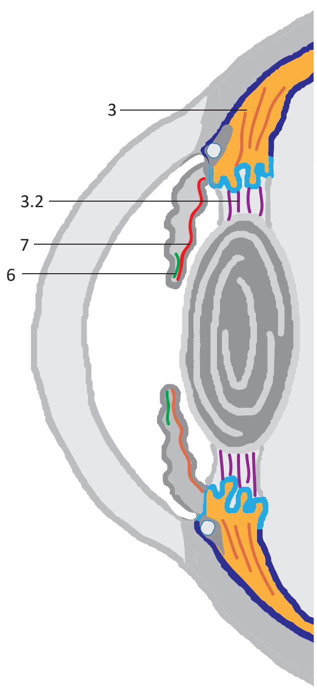
Vascular layer of eyeball (*tunica vasculosa bulbi / uvea*) – middle vascular layer

Choroid (*choroidea*)

– provides nutrition to the rods and cones of the retina
– the dark color of blood and the heavy pigmentation form a dark background for the visual receptors
– its elastic tension acts to slightly flatten the lens and so ensures that the lens can focus on distant objects in resting state
– **layers of the choroid:** suprachoroid lamina (*lamina suprachoroidea, lamina fusca sclerae*), perichoroidal space (*spatium perichoroideum*), vascular lamina (*lamina vasculosa*), capillary lamina (*lamina choroidocapillaris*), basal complex / Bruch's membrane (*complexus basalis*)

Ciliary body (*corpus ciliare*)

– produces aqueous humor, forms a muscular ring that is triangular in shape on cross-section
1 **Ciliary ring** (*orbiculus ciliaris / pars plana / anulus ciliaris*) – the outer non-folded pigmented part
2 **Corona ciliaris** / ciliary crown (*pars plicata*) – the inner folded part with processes
2.1 **Ciliary folds and processes** (*plicae et processus ciliares*) – enlarge the surface of the ciliary body
3 **Ciliary muscle** (*musculus ciliaris*) – smooth muscle
– its contraction relaxes the ciliary zonule, which causes the lens to thicken and focus on near objects
3.1 **Meridional, radial, circular and longitudinal fibres** (*fibrae meridionales, radiales, circulares, longitudinales*)
3.2 **Ciliary zonule of Zinn** (*zonula ciliaris*)
– a complex of the lenticular ligaments that attach to the ciliary body

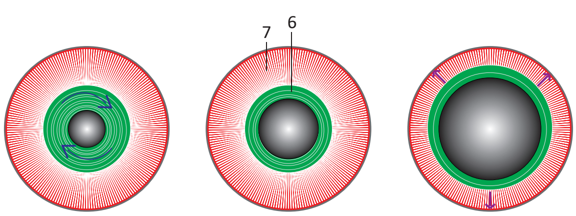


Vascular layer of the eyeball

Iris

– is a muscular ring that is narrow and rectangular in shape on cross-section
– is rich in melanocytes
– the number of melanocytes and amount of melanin determines the colour of the eyes
4 **Ciliary and pupillary margins** (*margo ciliaris et pupillaris*) – the outer and inner margins
5 **Greater and lesser rings of iris** (*anulus iridis major et minor*) – concentric circles on outer surface cover the dilator (greater) and sphincter (lesser) muscles
5.1 **Radial fold** (*plica radians*) – a serrate line on the anterior surface
6 **Sphincter pupillae** (*musculus sphincter pupillae*)
– smooth muscle that constricts the pupil, producing miosis
7 **Dilator pupillae** (*musculus dilatator pupillae*)
– smooth muscle that widens the pupil, producing mydriasis

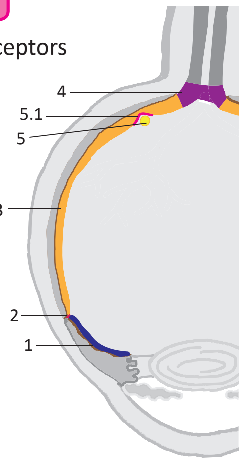
Pupil (*pupilla*) – a circular opening (2.5 – 4.5 mm) in the middle of the iris
1 **Miosis** – constriction of the pupil in reponse to light
– functions to protect the retina from excessive light
2 **Mydriasis** – dilation of the pupil in response to a lack of light
– increases the amount of light reaching the retina



Miosis Normal Mydriasis

Inner layer of eyeball (*tunica interna bulbi / retina*) – nervous layer

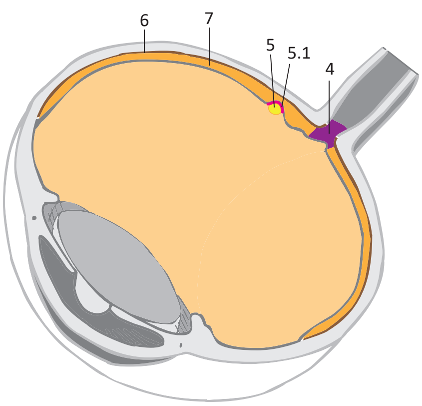
- 1 **Nonvisual retina** (*pars caeca*) – a blind part, not containing any photoreceptors
    - 1.1 **Ciliary and iridial parts of retina** (*pars ciliaris et iridica retinae*)
      – parts covering the ciliary body and iris
- 2 **Ora serrata** – the border between the nonvisual and optic part
- 3 **Optic part of retina** (*pars optica*) – is composed of 10 layers



Nervous layer of the eyeball

Optic part of the retina

- 4 **Optic disc** (*discus nervi optici*)
  – a spot without photoreceptors
  – the place where the optic nerve leaves the eyeball and the central retinal vessels (*arteria et vena centralis retinae*) enter
- 5 **Macula** (*macula lutea*)
  – the superficial layers of the retina divert around the macula
  – the spot of the sharpest visual acuity (macular vision)
  – located temporal (lateral) to the optic disc
    - 5.1 **Central fovea** (*fovea centralis*)
      – contains only cones and no rods
- 6 **Pigmented layer** (*stratum pigmentosum*)
- 7 **Neural layer** (*stratum nervosum*)
  – **photoreceptors – rods and cones** (*neura bacillifera et conifera*)
  – **transmitting neurons** (retinal bipolar and retinal ganglion cells)
  – **associative neurons (interneurons)** (horizontal and amacrine cells)
  – **supporting cells** (radial glial cells, radial cells of Müller)



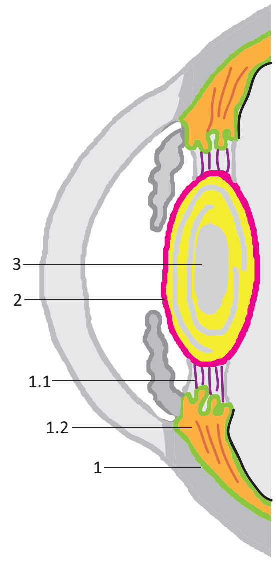
Nervous layer of the eyeball

Layers of the retina:

7.1 **Layer of rods and cones** (*stratum bacillorum conorumque*)
7.2 **Outer limiting membrane** (*stratum limitans externum*)
7.3 **Outer nuclear layer** (*stratum nucleare externum*)
7.4 **Outer plexiform layer** (*stratum plexiforme externum*)
7.5 **Inner nuclear layer** (*stratum nucleare internum*)
7.6 **Outer plexiform layer** (*stratum plexiforme internum*)
7.7 **Ganglionic layer** (*stratum ganglionicum*)
7.8 **Layer of nerve fibers** (*stratum neurofibrarum*)
7.9 **Inner limiting membrane** (*stratum limitans internum*)

Lens

- 1 **Ciliary body** (*corpus ciliare*)
  – holds the lens in position via the ciliary zonule
    - 1.1 **Ciliary zonule of Zinn** (*zonula ciliaris*)
      – the suspensory ligaments (apparatus) of the lens that attach it to the ciliary body
    - 1.2 **Ciliary muscle** (*musculus ciliaris*)
      – is responsible for lens accommodation
- 2 **Capsule of lens** (*capsula lentis*)
  – a transparent glassy layer of the lens
- 3 **Lens substance** (*substantia lentis*)
  – consists of the cortex and nucleus
- 4 **Anterior and posterior poles** (*polus anterior et posterior*)
- 5 **Axis and equator**



Lens and ciliary body

Vitreous body (*corpus vitreum*)

– transparent noncellular substance, is 98% water like a gel
– is located in the postremal (vitreous) chamber between the posterior surface of the lens and the retina
– is produced prenatally with no ability of regeneration
– when injured, it leaks out and is replaced with aqueous humour

The optic papilla *(papilla nervi optici)* is a clinical term for the optic disc.

Hyalos is the Greek term for the vitreous body.

The average value of intra-ocular pressure is 16 mmHg, but varies throughout the day from 10 to 21 mmHg. Daily production of aqueous humor is approximately 3 ml in each eyeball. The chambers of the eyeball contain approximately 0.2–0.3 ml of aqueous humor.

Measurements of the lens
Diameter: 9–10 mm
Thickness: 3.7–4.4 mm
Optic power: 15–20 dioptres

The cornea and lens are avascular. The eyeball possesses no lymph vessels. The lymph function is provided by the aqueous humor.

Clinical notes

Papilloedema means swelling of the optic disc. It is caused by increased intracranial pressure (intracranial hypertension) and is possible to see during examination of the fundus by ophthalmoscopy (aka fundoscopy).

The retina is not firmly attached to the choroid and thus it is liable to detachment *(amotio)*.

The retinal vessels and macula are prone to being affected by systemic disease such as diabetes and hypertension.

Lesions of branches of the central retinal artery can arise in hypertension, atherosclerosis and diabetes. They can cause impaired vision or even blindness.

Glaucoma is characterised by chronic and progressive atrophy of the optic nerve. It is usually caused by increased intraocular pressure.

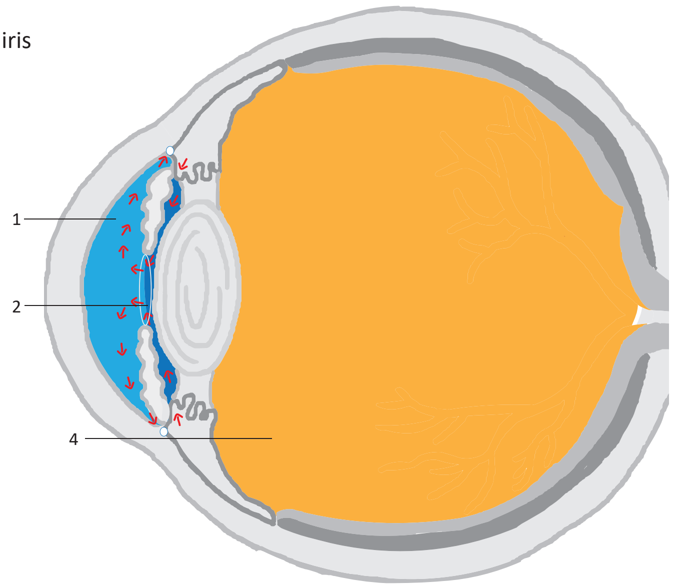
Cataract is a clouding (opacification) of the lens. The affected lens can be replaced by surgery. To replace the lens: an incision is made in the capsule of the lens, the opaque content is aspirated and an artificial lens is inserted into the lens.

Pseudophakia is a term for an artificial lens inserted into the eye.

Veins of the orbit have a different course than the arteries. They interconnect with the veins of the face and the intracranial venous sinuses. Such anastomoses allow spread of infection from the face into the brain via the orbit.

Chambers of eyeball (*camerae bulbi*)

- 1 **Anterior chamber** (*camera anterior*) – is located between the posterior surface of the cornea and the anterior surface of the iris
- 2 **Posterior chamber** (*camera posterior*)
  - – is located between the posterior surface of the iris, ciliary body and the anterior surface of the lens
  - – communicates with the anterior chamber through the pupil
- 3 **Aqueous humour** (*humor aquosus*)
  - – an intra-ocular transparent noncellular fluid
  - – nourishes the cornea, lens and vitreous body
  - **Production and flow of aqueous humour**
    1. produced in the ciliary body
    2. secreted into the posterior chamber
    3. passes through the pupil
    4. fills the anterior chamber
    5. enters the iridocorneal angle (*angulus iridocornealis*) and via the spaces of Fontana (*reticulum trabeculare*)
    6. is absorbed into the **canal of Schlemm** (scleral venous sinus, *sinus venosus sclereae*)
    7. drains through the aqeuous venules and then the scleral veins into the episcleral veins
- 4 **Postremal chamber / vitreous chamber** (*camera postrema / camera vitrea*)
  - – is located between the posterior surface of the lens and the retina
  - – contains the vitreous body



Chambers of the eyeball

Blood supply

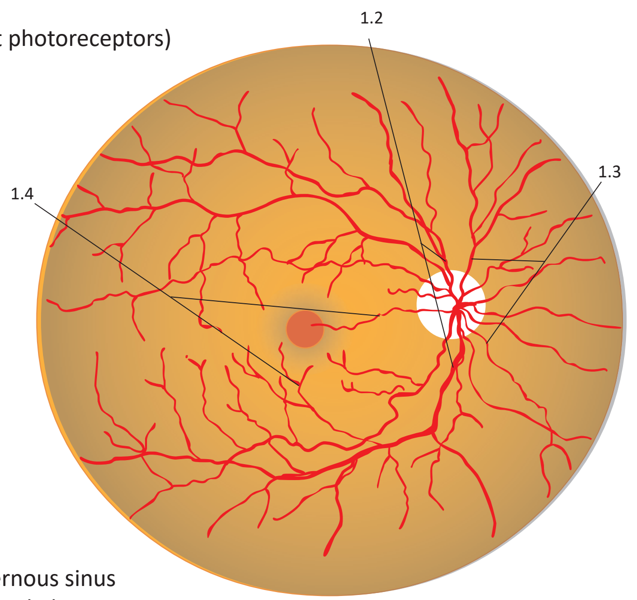
Arterial supply: ophthalmic artery (from the internal carotid artery)

1 **Central retinal artery** (*arteria centralis retinae*) – supplies the retina (except photoreceptors)
- 1.1 **Retinal blood vessels** (*vasa sanguinea retinae*)
  - – vessels of the eyeball fundus (*fundus oculi*)
- 1.2 Superior and inferior temporal retinal arterioles (*arteriola temporalis retinae superior et inferior*)
- 1.3 Superior and inferior nasal retinal arterioles (*arteriola nasalis retinae superior et inferior*)
- 1.4 Superior, middle and inferior macular arterioles (*arteriola macularis superior, media et inferior*)

2 **Lacrimal artery** (*arteria lacrimalis*)
- 2.1 Lateral palpebral arteries (*arteriae palpebrales laterales*)

3 **Short posterior ciliary arteries** (*arteriae ciliares posteriores breves*)
- – supply the choroid and photoreceptors
- 3.1 Cilioretinal artery (*arteria cilioretinalis*) (10–33 %)
  - – a variant accessory artery supplying the macula

4 **Long posterior ciliary arteries** (*arteriae ciliares posteriores longae*)
- – supply the ciliary body and iris



Arteries of the right ocular fundus

Venous drainage:

1 **Superior ophthalmic vein** (*vena ophthalmica superior*) – flows into the cavernous sinus
2 **Inferior ophthalmic vein** (*vena ophthalmica inferior*) – flows into the pterygoid plexus
3 **Angular vein** (*vena angularis*) – into the facial vein

Veins of the eyeball correspond to the arteries excluding:

1 **Scleral veins**
2 **Canal of Schlemm / scleral venous sinus** (*sinus venosus sclereae*) – flows into the scleral veins and then to the anterior ciliary veins
3 **Vorticose veins** (*venae vorticosae*) – usually four in the four quadrants of the eyeball, collect blood from the choroid

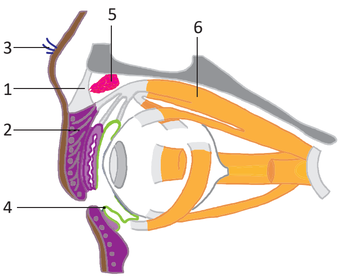
Innervation

Special sensory: the optic nerve terminates lateral geniculate body of the diencephalon
– the visual pathway continues to cortical area 17 in the occipital lobe
Somatosensory: ophthalmic nerve (first branch of the trigeminal nerve)
– **cornea:** long ciliary nerves
– **surrounding area:** lacrimar nerve, frontal nerve, nasociliary nerve and infraorbital nerve
Sympathetic innervation of the dilatator pupillae: short ciliary nerves (arising from the ciliary ganglion), postganglionic sympathetic fibers originates in the superior cervical ganglion (do not interconnect in the ciliary ganglion in comparison with the parasympathetic fibers)
Parasympathetic innervation of the sphincter pupillae and ciliary muscle: short ciliary nerves, which synapse at the ciliary ganglion (preganglionic fibres form a component of the inferior branch of the oculomotor nerve)

The conjunctiva covers the sclera on the external part of the eyeball (the white of the eye). It is reflected onto the internal surface of the eyelids and thus **allows smooth eyelid movements on surface of the eyeball**. The conjunctiva and lacrimal apparatus form an **important part of the immune system of the eye** as they contain extensive lymph tissue and produce mucus. **The lacrimal glands produce tears**, which clean the outer surface of the eyeball. Tears are drained via the nasolacrimal duct into the nasal cavity. **The extra-ocular muscles** are striated muscles that produce very fine eye movements.

Parts

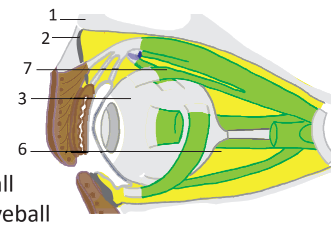
- 1 **Ligamentous apparatus** (*apparatus ligamentosus*)
- 2 **Eyelids** (*palpebrae*)
- 3 **Eyebrow** (*supercilium*)
- 4 **Conjunctiva** (*tunica conjunctiva*)
- 5 **Lacrimal apparatus** (*apparatus lacrimalis*)
- 6 **Muscular apparatus** (*apparatus muscularis*)



Structures of the orbit

Ligamentous apparatus (*apparatus ligamentosus*)

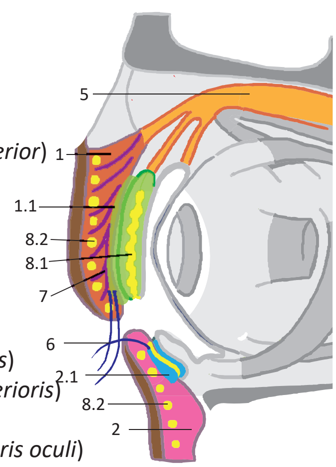
- 1 **Periorbit** (*periorbita*) – the periosteum of the orbit
- 2 **Orbital septum** (*septum orbitale*) – a fibrous plate enclosing the orbit ventrally and extending into the eyelids
- 3 **Fascial sheath of eyeball / capsule of Tenon** (*vagina bulbi*) – a fibrous cover of the eyeball
- 4 **Suspensory ligament of eyeball** (*lig. suspensorium bulbi*) – a caudal strengthened band of the fascial sheath of the eyeball
- 5 **Episcleral space** (*spatium episclerale*) – a space between the eyeball and its fascial sheath filled with loose connective tissue
- 6 **Retrobulbar fat / orbital fat body** (*corpus adiposum orbitae*) – a fat pad inside the orbit
- 7 **Muscular fasciae** (*fasciae musculares*) – fasciae of the extra-ocular muscles



Ligamentous apparatus

Eyelids (*palpebrae*)

- 1 **Superior eyelid / upper eyelid** (*palpebra superior*)
  - 1.1 **Superior tarsus** (*tarsus superior*) – a thick (10 mm) plate of connective tissue in the upper eyelid
  - 1.2 **Superior tarsal muscle of Müller** (*musculus tarsalis superior*)
- 2 **Inferior eyelid / lower eyelid** (*palpebra inferior*)
  - 2.1 **Inferior tarsus** (*tarsus inferior*) – a thick (5 mm) plate of connective tissue in the lower eyelid
  - 2.2 **Inferior tarsal muscle** (*musculus tarsalis inferior*)
- 3 **Medial and lateral palpebral ligaments** (*ligamentum palpebrale mediale et laterale*)
- 4 **Lateral and medial eye angle** (*angulus oculi medialis et lateralis*)
- 5 **Levator palpebrae superioris** (*musculus levator palpebrae superioris*)
- 6 **Eyelashes** (*cilia*)
- 7 **Palpebral part of orbicularis oculi** (*pars palpebralis m. orbicularis oculi*)
- 8 **Glands** – three types of glands
  - 8.1 **Tarsal glands of Meibom** (*glandulae tarsales*) – holocrine sebaceous glands
  - 8.2 **Ciliary glands** (*glandulae ciliares*) – apocrine glands
  - 8.3 **Sebaceous gland of Zeis** (*glandulae sebaceae*) – holocrine sebaceous glands

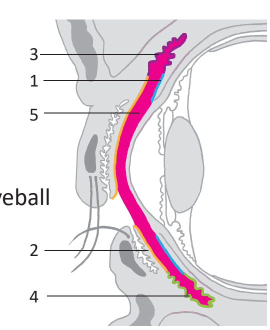


Eyelids

Conjunctiva (*tunica conjunctiva*)

– is composed of stratified columnar epithelium
– contains goblet cells, large amount of lymphoid tissue and the conjunctival glands of Wolfring (*glandulae conjunctivales*)

- 1 **Bulbar conjunctiva** (*tunica conjunctiva bulbi*) – converts into the corneal epithelium on the ventral surface of the eyeball
- 2 **Palpebral conjunctiva** (*tunica conjunctiva palpebrarum*) – the conjunctiva on the dorsal surface of the eyelid
- 3 **Superior conjunctival fornix** (*fornix conjunctivae superior*) – contains the openings of the lacrimal gland ducts
- 4 **Inferior conjunctival fornix** (*fornix conjunctivae inferior*)
- 5 **Conjunctival sac** (*saccus conjunctivalis*) – a space between the conjunctiva and cornea



Conjunctiva

Blefaron is the Greek term for eyelid.

The orbitalis / orbital muscle *(musculus orbitalis)* is a smooth muscle located in the inferior orbital fissure.

Terminology of directions of eye movements
Nasal (medial): towards the nose
Temporal (lateral): towards the temporal region

Saccadic eye movements can be observed when reading or when shifting focus from one object to another.

Rapid eyes movements characterise the rapid eye movement phase (REM) of the sleep cycle.

The suspensory ligament of the eyeball holds the eyeball as if it was in a hammock.

The palpebral part of the orbicularis oculi is innervated by the facial nerve.

The superior and inferior tarsal muscles are smooth muscles.

Clinical notes

Chalazion is inflammation of the tarsal glands. It presents as a yellow and often painful lump on the internal surface of the eyelid.

Hordeolum is inflammation of the outer eyelid glands located at the eyelid margin.

Palsy of the facial nerve manifests as an impairment in function of the orbicularis oculi, which causes an inability to close the palpebral fissure (lagophthalmus). This can lead to desiccation of the conjunctiva (xerophthalmia), which can gravely endanger the acuity of vision by predisposition to inflammation and ulceration of the cornea.

Conjunctivitis is inflammation of the conjunctiva. It can have allergic and infectious aetiologies.

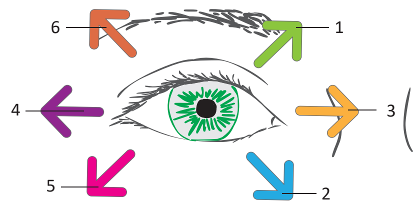
Epiphora is an overflow of tears onto the face.

Strabism / squint / heterotropia / strabismus is a symptom of extra-ocular muscle palsy.

Muscular apparatus (*apparatus muscularis*)

Extra-ocular muscles / extrinsic muscles of the eyeball (*musculi externi bulbi oculi*)

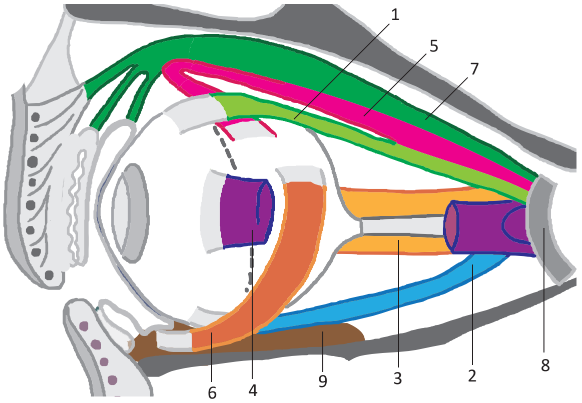
- 1 **Superior rectus** (*musculus rectus superior*) – nasocranial movement
- 2 **Inferior rectus** (*musculus rectus inferior*) – nasocaudal movement
- 3 **Medial rectus** (*musculus rectus medialis*) – nasal movement
- 4 **Lateral rectus** (*musculus rectus lateralis*) – temporal movement
- 5 **Superior oblique** (*musculus obliquus superior*) – temporocaudal movement
  – its tendon winds around the cartilaginous trochlea (trochlear spine, *spina trochlearis*)
- 6 **Inferior oblique** (*musculus obliquus inferior*) – tempocranial movement
- 7 **Levator palpebrae superioris** (*musculus levator palpebrae superioris*) – elevates the upper lid
  – located above the superior rectus and inserts in the superior tarsus



Directions of eye movements controlled by the extra-ocular muscles

Muscle origins

- 8 **Common tendinous ring of Zinn** (*anulus tendineus communis*)
  – the common origin of the rectus muscles, superior oblique muscles and levator palpebrae superioris
- 9 **Inferior wall of the orbit, medially**
  – is located lateral to the posterior lacrimal crest
  – the origin of the inferior oblique



Muscle insertions:

1 **Insertions ventral to the eyeball equator** (rectus muscles)
– move the eyeball in the direction of the muscle insertion

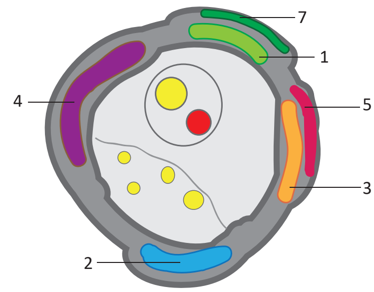
2 **Insertions dorsal to the eyeball equator** (obliquus muscles)
– move the eyeball opposite to the direction of the muscle insertion

Muscle innervation

III **Oculomotor nerve** – superior, inferior and medial rectus, inferior oblique, levator palpebrae superioris

IV **Trochlear nerve** – superior oblique

VI **Abducens nerve** – lateral rectus



Common tendinous ring, right orbit

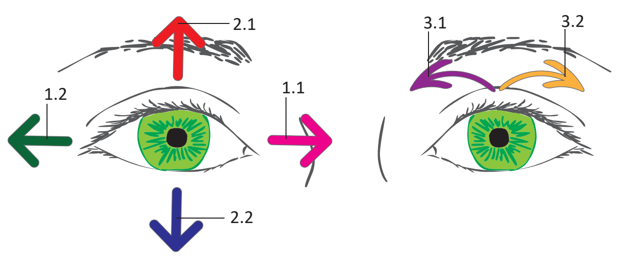
Eyeball movements

Movements of one eyeball (individual):

1 **Around the vertical axis:**
- 1.1 **Adduction** – nasal movement
- 1.2 **Abduction** – temporal movement

2 **Around the horizontal axis:**
- 2.1 **Elevation** – cranial movement
- 2.2 **Depression** – caudal movement

3 **Around the center:**
- 3.1 **Intorsion** – nasal rotation of the superior eyeball margin
- 3.2 **Extorsion** – temporal rotation of the superior eyeball margin



Combined eye movements

4 **Version** – both eyes moving synchronously and symmetrically in the same direction

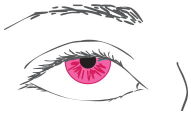
5 **Vergence** – simultaneous movements of both eyes in the opposite direction

5.1 **Divergence** – simultaneous outward movement of both eyes away from each other

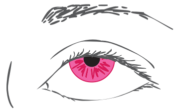
5.2 **Convergence** – simultaneous inward movement of both eyes toward each other

Eye divergence

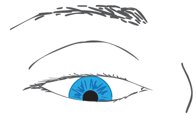
Eye convergence



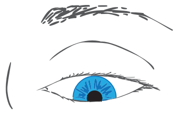
Eye elevation
superior rectus
inferior oblique



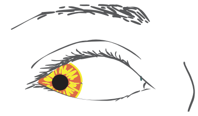
Eye elevation
superior rectus
inferior oblique



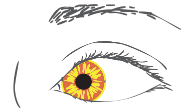
Eye depression
inferior rectus
superior oblique



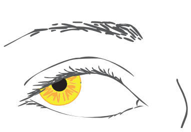
Eye depression
inferior rectus
superior oblique



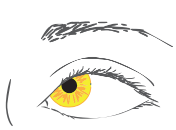
Eye abduction
lateral rectus



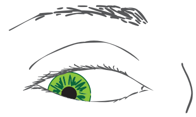
Eye adduction
medial rectus



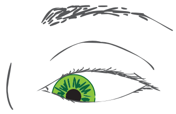
Eye elevation and abduction
inferior oblique



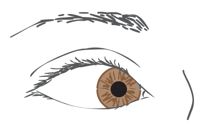
Eye elevation and abduction
superior rectus



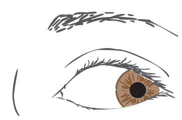
Eye depression and abduction
superior oblique



Eye depression and abduction
inferior rectus



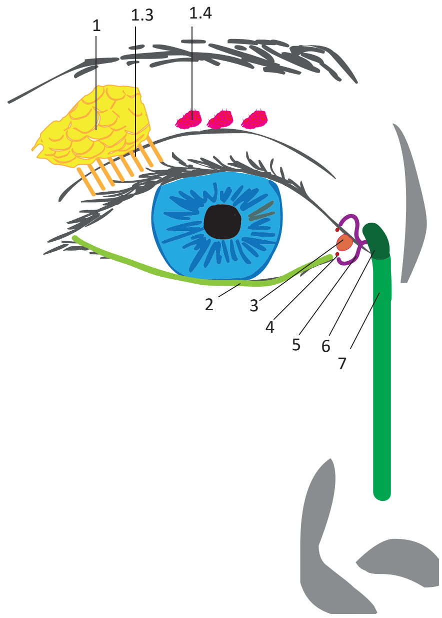
Eye adduction
medial rectus



Eye abduction
lateral rectus

Lacrimal apparatus (*apparatus lacrimalis*)

- 1 **Lacrimal gland** (*glandula lacrimalis*) – a tuboalveolar serous gland with myoepithelial cells
  – divided into two parts by the course of the levator palpebrae superioris
  - 1.1 **Orbital part** (*pars orbitalis*)
  - 1.2 **Palpebral part** (*pars palpebralis*)
  - 1.3 **Excretory ducts** (*ductuli excretorii*)
    – 12 to 15 individual ducts
  - 1.4 **Accessory lacrimal ducts of Krause** (*glandulae lacrimales accessoriae*)
- 2 **Lacrimal pathway** (*rivus lacrimalis*)
- 3 **Lacus lacrimalis / lacrimal lake** (*lacus lacrimalis*)
  - 3.1 **Lacrimal caruncle** (*caruncula lacrimalis*)
    – an elevation of the conjunctiva at the medial angle of the eye
  - 3.2 **Lacrimal papilla** (*papilla lacrimalis*)
    – an elevation at the palpebral margin
- 4 **Superior and inferior lacrimal punctum** (*puctum lacrimale superius et inferius*)
  – origins of the lacrimal canaliculi
- 5 **Superior and inferior lacrimal canaliculus** (*canaliculus lacrimalis superior et inferior*)
  – canals draining tears into the lacrimal sac
- 6 **Lacrimal sac** (*saccus lacrimalis*)
  – canaliculi join and open into the lacrimal sac
- 7 **Nasolacrimal duct** (*ductus nasolacrimalis*)
  – is located inside the bony nasolacrimal canal
  – opens into the inferior nasal meatus
  - 7.1 **Lacrimal fold of Hasner** (*plica lacrimalis*)
  - 7.2 **Inferior nasal meatus of nasal cavity** (*meatus nasi inferior cavitatis nasi*)



Lacrimal apparatus of the right eye

1.6 **Touch, pain, proprioception** – Tactus, nociceptio, proprioceptio

It is important to classify the **sense of touch** into two point discrimination, tension, pressure and vibrations and to differentiate it from the **sense of pain** (nociception). Both senses are perceived by identical receptors. **Perception depends on the intensity of stimulation**.

Exteroceptors

1 **Epidermis**
- 1.1 **Free nerve ending** (*terminatio neuralis libera*)
- 1.2 **Tactile meniscus of Merkel** (*meniscus tactilis*)

2 **Dermis**
- 2.1 **Tactile corpuscle / Meissner's corpuscle** (*corpusculum ovoideum*)
- 2.2 **Bulbous corpuscle / Ruffini corpuscle** (*corpusculum sensorium tactile*)

3 **Subcutaneous tissue** (*hypodermis*)
- 3.1 **Lamellar corpuscle / Pacinian corpuscle** (*corpusculum lamellosum*)

4 **Interoceptors**
- 4.1 **Free nerve ending** (*terminatio neuralis libera*)

5 **Proprioceptors**
- 5.1 **Muscle spindle** (*fusus neuromuscularis*)
  – situated in striated muscles (few in the extra-ocular muscles, none in the lingual muscles)
  – composed by a fibrous capsule and intrafusal muscular fibers
  – innervated by γ-motoneurons (γ-loops)
  – sensory anulospiral (center) and flower spray ending (periphery)
- 5.2 **Golgi tendon organ** (*organum sensorium tendinis*)

Senses and skin

Dakryon is the Greek term for tear.

The tear film *(irroratio lacrimarum)* is a thin layer of liquid covering the conjunctiva and cornea. A lack of tear film is found in the dry eye syndrome.

Lacrima (tear) is a watery fluid.

In newborns, the lacrimal canaliculi may not be patent until the 3rd month after birth. An obstruction of the nasolacrimal duct causes epiphora. The excretory lacrimal ducts contain small folds (plicae), sometimes wrongly termed valvulae.

Cutis is the Latin term for skin. **Derma** is the Greek term.

Skin thickness: 0.5–4 mm

Skin types according to structure and location:
Thin skin (hairy skin) lacks the stratum lucidum. The epidermis is 75–150 µm thick.
Thick skin (smooth non-hairy skin) is found on the palms and soles. The epidermis is 400–800 µm thick.

Dermal ridges and skin sulci in the palms and soles are arranged in longitudinal papillary lines. In fingertips, they form unique patterns called dermatoglyphs and are implicated in dactyloscopy (fingerprint indentification).

The whole epidermis regenerates every 15–30 days (skin of the back every 28 days and of the head every 14 days). Acceleration of epidermal cell proliferation occurs in a disease called psoriasis vulgaris.

Cutaneous striae *(striae cutaneae, striae distensae)* are tears in the reticular layer of the dermis, which typically occur during growth, gravidity and obesity.

Clinical notes

During surgical operations, it is useful to direct incisions along the tension lines. This provides the optimal functional and cosmetic effects.

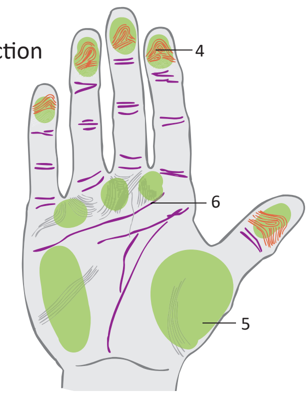
The direction of collagen fibres determins the relaxed skin tension lines of Langer (RSTL). Making incisions along these lines minimises the extent of enlargement of the surgical wound by skin tension.

Lines of maximal extensibility (LME) are perpendicular to the relaxed skin tension lines. They show the direction the skin can be maximally stretched. This is useful when planning skin flaps.

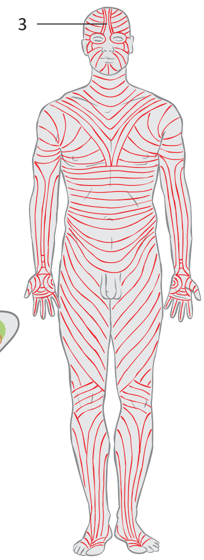
The skin is the **largest organ** of the human body. Its function are **protection against the external environment**, **thermoregulation** (via its the sweat glands and regulation of its blood flow), **excretion** (sweating and insensible water loss), **absorption** (of medications), **immunity**, **metabolism** (vitamin D metabolism from ergosterol) and **expression of emotions and mental states** (via vasoactivity and sweat production). It is composed of the **epidermis** and **dermis** and is continuous with the underlying **hypodermis** (subcutaneous tissue). The appendages of the skin include the hair, nails and skin glands.

Macroscopic features of the skin

1 **Skin sulci** (*sulci cutis*) – skin grooves bordering rhomboid skin areas
2 **Skin areas** (*areae cutis*) – are formed by traction of the elastic and collagen fibres of the dermis
– correspond to the tension lines
3 **Tension lines / cleavage lines** (*lineae distractiones*) – are located in the direction of the dermal fibril bundles and are perpendicular to the direction of the highest traction
– correspond to wrinkles on the surface of the skin
4 **Dermal ridges / papillar ridges** (*cristae cutis*) – tactile ridges on the palm and sole
5 **Tactile elevations** (*roruli tactiles*)
– elevated areas in the hand that are rich in sensory nerve endings
– 1 thenar, 1 hypothenar, 3 interdigital and 5 terminal tactile elevations
6 **Tension lines of Langer / cleavage lines** (*lineae distractionis*)
– flexion grooves at joints, on the palms and wrinkles
– include the mentolabial, nasolabial, suprapalpebral, infrapalpebral, gluteal and inguinal grooves, genitofemoral and intergluteal (natal) groove



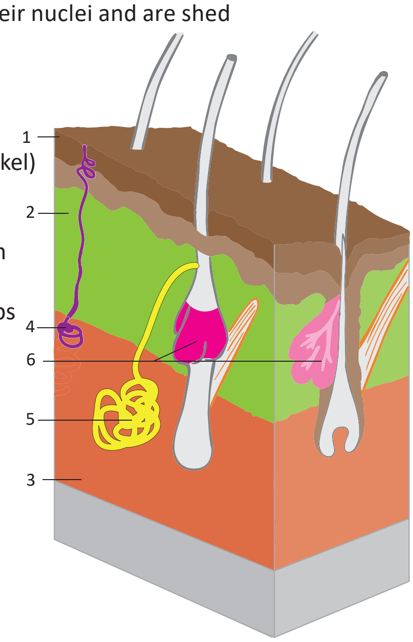
Tactile pads within the hand



Tension lines

Histological structures

1 **Epidermis**
1.1 **Layers:** stratum basale (basal cells layer), st. spinosum, st. granulosum, st. lucidum, st. corneum, st. disjunctum
1.2 **Cells:**
1.2.1 **Keratinocytes** – migrate from the stratum basale to the surface, where they lose their nuclei and are shed
1.2.2 **Melanocytes** – cells of neuroectodermal origin that produce melanin, which protects agains UV radiation (cells transfer their melanin to keratinocytes)
1.2.3 **Langerhans cells** – dendritic cells of the immune system
– originate in the bone marrow and present antigens to lymphocytes
1.2.4 **Merkel cells** – cells of neuroectodermal origin, posses tactile receptors (discs of Merkel)
2 **Dermis** (*corium*) – is composed of connective tissue and contains capillaries, skin appendages and tactile receptors
2.1 **Papillary layer** (*stratum papillare*) – is composed of collagenous and elastic fibres that form the dermal papillae that project into the epidermis forming the epidermal ridges
2.2 **Reticular layer** (*stratum reticulare*) – is composed of collagen fibres forming rhomboid loops
3 **Subcutaneous tissue** (*hypodermis / tela subcutanea*) **–** is composed mainly of adipose tissue with a variable thickness (thickest in the abdomen, buttocks and thigh)
3.1 **Fatty layer** (*panniculus adiposus*)
3.2 **Fibrous layer** (*stratum fibrosum*)
3.3 **Membranous layer** (*stratum membranosum*) – fibrous thickenings
3.4 **Loose connective tissue** (*textus connectivus laxus*)
3.5 **Skin ligaments** (*retinacula cutis*)
3.6 **Subcutaneous synovial bursae** (*bursae synoviales subcutaneae*)



Section of the skin

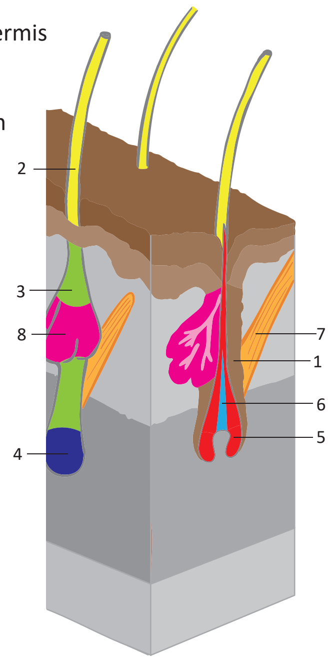
Skin glands

4 **Eccrine sweat glands** (*glandulae sudoriferae merocrinae/eccrinae*)
– simple, coiled, tubulous, merocrine glands that secrete the fluid outside of hair follicles
– are located everywhere except the lips, internal surface of the prepuce, glans penis and nail beds, are in the palm, sole and forehead
– secrete sweat (sudor) – play a role in thermoregulation and are innervated by cholinergic nerve fibres
5 **Apocrine sweat glands** (*glandulae sudoriferae apocrinae*) – become completely functional at puberty (controlled by androgens)
– open into hair follicles, are located in the axilla, perineal region and labia minora
– produce a viscous secretion without any smell (the aromatic smell is caused by bacteria), are innervated by adrenergic fibres
5.1 **Mammary gland** (*glandula mammaria*), **areolar glands of Montgomery** (*glandulae areolares*)
5.2 **Ciliary glands of Moll** (*glandulae ciliares*) – located in the eyelids
5.3 **Ceruminous glands** (*glandulae ceruminosae*) – located in the external acoustic meatus (producing cerumen)
5.4 **Nasal sweat glands** (*glandulae sudoriferae nasales*) – located in the nasal vestibule
5.5 **Circumanal glands** (*glandulae circumanales*) – located around the anus
6 **Sebaceous glands** (*glandulae sebaceae*) – holocrine, alveolar glands, produce the sebum
– situated everywhere except the palms and soles
– open into hair follicles, independently on the hair follicles on the lips, glans penis, prepuce, labia minora, eyelids

Appendages of the skin

Hairs (*pili*)

- 1 **Follicle** (*foliculus*) – a hair sheath formed by invaginated epidermis
- 2 **Hair shaft** (*stipes/scapus*)
- 3 **Hair root** (*radix*)
- 4 **Hair bulb** (*bulbus*) – an derivative of the stratum germinativum that is composed of replicating cells
  – the dermal papilla / papilla of the hair follicle (*papilla dermalis pili*) is situated at the base of the hair bulb
- 5 **Cortex**
- 6 **Medulla**
- 7 **Arrector pili muscle** (*musculus arrector pili*)
  – smooth muscle that inserts into the hair follicle
  – functions to erects the hair
- 8 **Sebaceous glands** (*glandulae sebaceae*)
  – open into the hair follicles
  – produce sebum, an oily secretion, that waterproofs the skin



Section of the skin

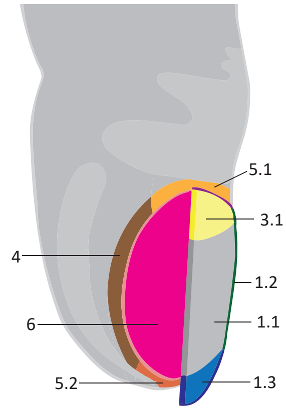
Types of hair:
1 **Primary hair** – disappears before birth
– appears between the 4th and 6th months of intrauterine life
1.1 **Downy hair** (*lanugo*)
2 **Secondary hair**
2.1 **Hairs** (*pili*) – replaces downy hair
2.2 **Scalp hairs** (*capilli*)
2.3 **Eyebrows** (*supercilia*)
2.4 **Eyelashes** (*cilia*) – form one row on the lower eyelid, but multiple rows on the upper eyelid
3 **Tertiary hair** – appear in puberty (dark stiff hairs)
3.1 **Axillary hairs** (*hirci*) – is situated in the axilla
3.2 **Pubic hairs** (*pubes*) – is situated in the pubic area
3.3 **Beard** (*barba*) – is situated on the face
3.4 **Hairs of tragus** (*tragi*) – is situated in the external meatus
3.5 **Hairs of vestibule of nose** (*vibrissae*)
– situated in the nasal vestibule
3.6 **Sinuous hairs** – long tactile hairs in the eyebrows, face, forearm and wrist (appear after the 30th year of age)

Nail (*unguis*)

1 **Body** (*corpus*)
- 1.1 **Nail plate** (*lamina unguis*)
- 1.2 **Lateral border** (*margo lateralis*)
- 1.3 **Free border** (*margo liber*) – the distal margin

2 **Nail root** (*radix unguis*)
3 **Nail matrix** (*matrix unguis*) – nail grows from the matrix
- 3.1 **Lunule** (*lunula*) – the part of the nail matrix shining through the nail plate
- 3.2 **Hidden border** (*margo occultus*) – the proximal border at the nail matrix

- 4 **Proximal and lateral nail fold** (*vallum proximale et laterale*) – skin folds surrounding the nail
- 5 **Perionyx** – the skin surrounding the nail
  - 5.1 **Eponychium** – the skin proximally covering the nail matrix
  - 5.2 **Hyponychium** – the skin distally under the nail free margin
- 6 **Nail bed** (*lectulus*) – the dermal base of the nail
  – capillaries shine through the nail bed ("a window for blood oxygenation")



Parts of the nail

Blood supply

Arterial supply: at the boundary of the subcutaneous tissue and dermis: cutaneous plexus
within the dermis: subpapillary plexus (*plexus subpapillaris*)
Venous drainage: cutaneous plexus, subpapillary plexus
Lymphatic drainage: superficial (subpapillary) and deep cutaneous network

Innervation

Somatosensory (touch and pain): free nerve endings and corpuscles
Sympathetic: for vessels, it is represented by adrenergic postganglinic vasomotor fibers
– **for sweat glands** by cholinergic nerve fibers

The angiosome is an area tissue supplied by one vessel. Angiosomes are interconnected by arteries of small caliber or by arterio-arterial anastomoses.

Hairs are present everywhere on the skin except on the palms, soles, lips, glans penis, glans clitoridis and labia minora. Hair grows at a speed about 0.4 mm per day. About 100 hairs are physiologically shed per day.

Onychos is the Greek term for nail.

The thickness of the nail is 0.5–1.5 mm. Nails grow at a speed of 0.1 mm per day.

Breast size in Europe is given by the circumference of the thorax just below the breast in the inframammary fold and by a numeric code determining the difference between the circumference measured over the nipple and the circumference below the breast (AA-I).

Mastos is the Greek term for breast. **Mammilla** is the clinically term for nipple.

Clinical notes

Knowledge of the blood supply and innervations of the skin is essential for skin flap surgery. A skin flap surgery is a transfer of certain skin areas to cover defects.

Alopecia means a loss of hair.

It is necessary to remove the axillary lymph nodes when performing mastectomy for advanced breast tumours. Only level 1 and level 2 nodes (according to Berg's classification) are dissected. Level 3 nodes are left as they lie close to the axillary neurovascular bundle and so carry a high risk of complications such as lymphoedema of the upper extremity.

The insertion of the suspensory ligaments (retinaculum) of the breast to the skin gives an explanation as to why skin sometimes retracts over breast tumours.

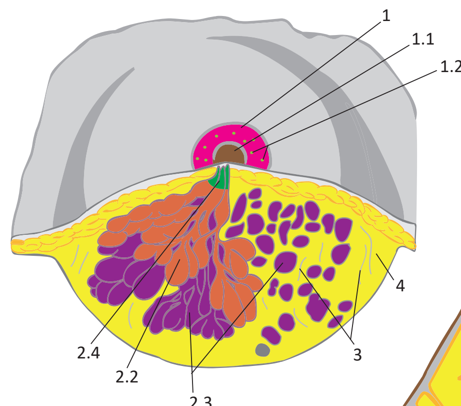
Accessory breast tissue can be found at any point along the course of the mammary crest. This condition is known as polymastia. The most common form accessory breast tissue takes is a supernumery nipple (polythelia).

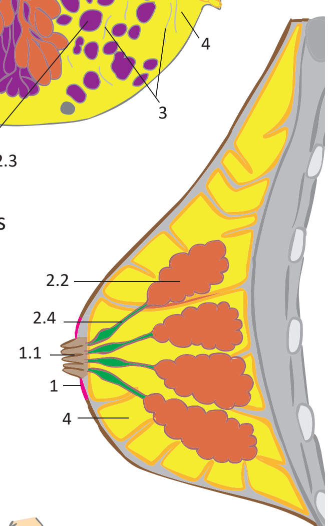
The areolomammilar complex is a term for the areola and nipple. Sensory nerves approach from above, which is why surgical incisions are performed below the lower border of the areola.

The breast is a paired organ located on the thorax. **The areola** is a pigmented area of skin located at the apex of the breast. The **nipple is located in the centre of the areola** and serves as a opening of the lactiferous ducts. The mammary gland and **fat tissue** are located inside the breast. The mammary gland is the **largest skin gland** and **produces milk for the newborn**. The mammary gland is derived from the mammary crest in both sexes and is located between the 3rd and 6th intercostal spaces. The weight of the breast increases during puberty and lactation. The usual of the breast is 150 g but can increase to up to 300–800 g during lactation .

Structure

- 1 **Areola** (*areola mammae*) – a circular pigmented area around the nipple
  - 1.1 **Nipple** (*papilla mammaria*) – contains the openings of the lactifeours ducts
    – contains smooth muscle responsible for its erectility
  - 1.2 **Areolar tubercles / tubercles of Montgomery** (*tubercula areolaria*)
    – small elevation made by the areolar glands
  - 1.3 **Areolar glands / glands of Montgomery** (*glandulae areolares*)
    – small apocrine glands
- 2 **Mammary gland** (*glandula mammaria*)
  – a compound tuboalveolar apocrine gland
  - 2.1 **Axillary process** (*processus axillaris*) – a gland projection towards the axilla
  - 2.2 **Lobes of mammary gland** (*lobi glandulae mammariae*)
    – 15–20 lobes that are subdivided into lobules
  - 2.3 **Lobules of mammary gland** (*lobuli glandulae mammariae*)
    – lobules composed of alveoli
  - 2.4 **Lactiferous ducts** (*ductus lactiferi*) – ducts of the mammary gland
    - 2.4.1 **Lactiferous sinuses** (*sinus lactiferi*) – the widened terminal portions of the lactiferous ducts
- 3 **Suspensory ligaments of breast / Cooper's ligament**
  (*ligamenta suspensoria mammae / retinaculum cutis mammae*)
  – a fibrous network inserting into the pectoral fascia and dermis of the skin over the breast
- 4 **Adipose layer / mammary fat pad** (*stratum adiposum mammae / corpus adiposum mammae*)
  – is located both in front and behind the gland
  – the volume of fat determines the size of the breast

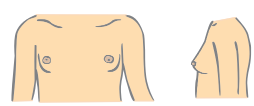


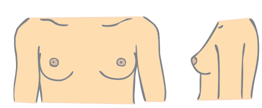


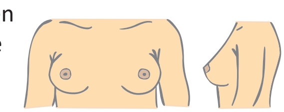
Sagittal section of the breast

Breast development and shapes

1. **Mammary crest** *(crista mammaria)*
   – is formed within the 1st month of intrauterine life
   – extends from the anterior axillary line down to just above the groin
2. **Mammary gland base** – is present at birth
3. **Breast enlargement during puberty** – is classified by Tanner stages M1 to M5
   3.1 **Mamma puerilis/neutralis**
   3.2 **Mamma areolata**
4. **Adult mamma**
   4.1 **Mamma disciformis** – small flat breast
   4.2 **Mamma hemispheroidea** – hemispherical breast
   4.3 **Mamma piriformis** – pear-shaped breast that is typical for middle-aged women
   4.4 **Mamma pendula** – flabby breast of elderly women caused by loss of fat tissue


Mamma disciformis

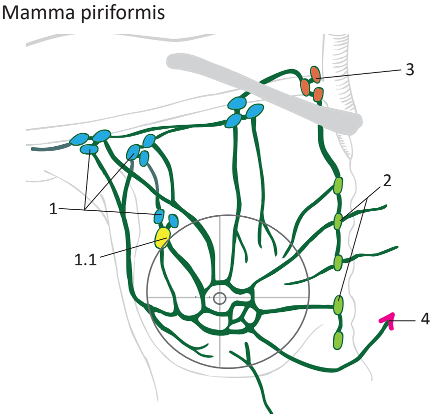

Mamma hemispheroidea


Mamma piriformis

Blood supply and innervation

Arterial supply: mammary branches (rami mammarii)
– internal, superior and lateral thoracic artery, intercostal arteries (2–4)
Venous drainage: internal and lateral thoracic vein
Lymphatic drainage: starts as the subareolar plexus of Sappey
- 1 **Axillary lymph nodes**
  - 1.1 **Node of Sorgius** – is a sentinel lymph node of the mammary gland
    – is the most caudal of the pectoral lymph nodes
    – is located at the 2nd/3rd origin of the serratus anterior
- 2 **Parasternal lymph nodes**
- 3 **Supraclavicular lymph nodes**
- 4 **Contralateral axillary lymph nodes**

Sensory innervation: intercostal nerves (4–6) – nipple (dermatome T4)
Autonomic innervation: periarterial plexuses



Lymphatic drainage of the breast

I. Senses (in general) and the internal environment

1. State the receptors according to their individual stimuli. (p. 502)
2. Explain how receptors monitor the internal environment. (p. 502)
3. List the 6 individual senses and sensory organs of their perception. (p. 502)
4. State the chemoreceptors location in the body and specify their function. (p. 502)
5. Clarify the difference between the carotid and para-aortic body. (p. 502)
6. Describe and give the locations of the receptors within the cerebral ventricles walls. (p. 502)

II. Olfactory organ

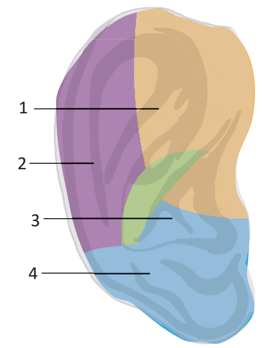
7. Describe the olfactory organ. (p. 503)
8. State the pathway of olfactory cells axons to the brain. (p. 503)
9. Explain why the olfactory mucosa includes glands. (p. 503)

III. Gustatory organ

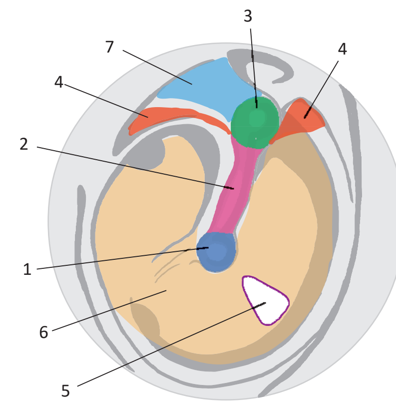
10. Describe the localization of the taste buds. (p. 503)
11. Classify the 5 types of taste. (p. 503)
12. State three nerves involved in the transmission of gustatory stimuli and specify which parts of the tongue they innervate. (p. 503)

IV. The organs of audition and balance

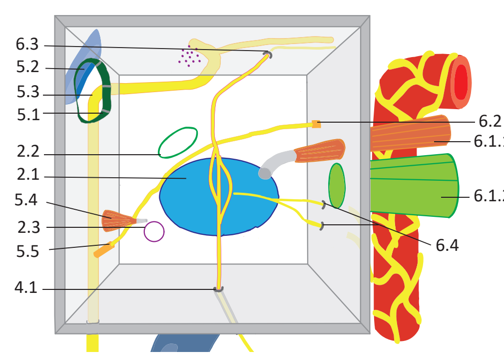
13. State two functions of the ear. (p. 504)
14. Specify the type of cartilage found in the auricle. (p. 505)
15. Describe the structural base of the umbo of the tympanic membrane. (p. 505)
16. Describe the anatomy of the auditory ossicles in the middle ear. (p. 506)
17. Explain the function of the auditory tube. (p. 506)
18. Describe the muscles of the tympanic cavity and state their innervations. (p. 506)
19. State the two openings in the vestibule of the osseous labyrinth. (p. 507)
20. Explain the different function of the utricle and saccule. (p. 507)
21. State the 5 structures passing through the internal acoustic meatus. (p. 507)
22. Define the term otoconia. (p. 507)



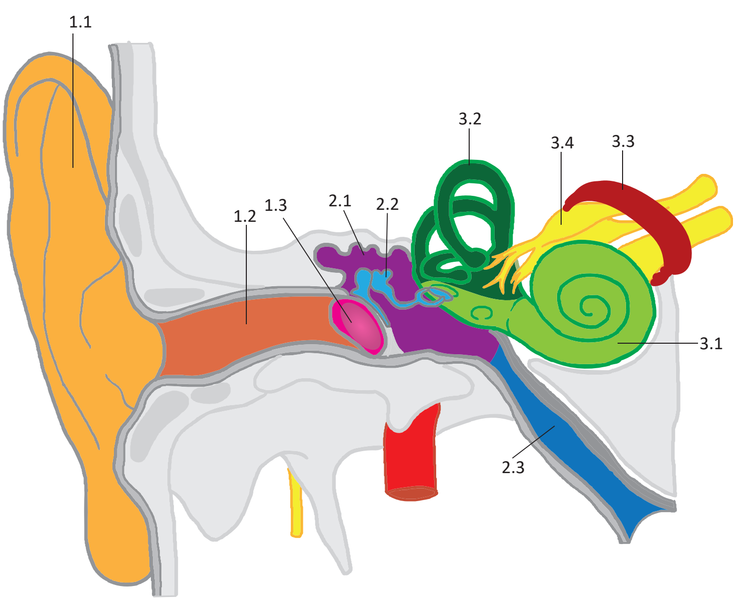
Describe the somatosensory innervation of the auricle



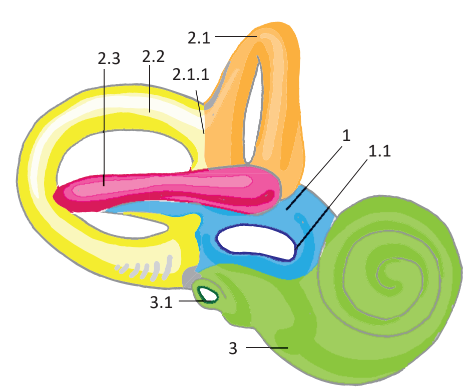
Describe the structure of the tympanic membrane from otoscopic view



Describe the structures within the tympanic cavity



Describe the structure of the ear



Describe the structure of the bony labyrinth

V. Visual organ

23. Describe the eyeball structure (using geographical terminology). (p. 508)
24. Describe the difference between the optic and visual axes. (p. 508)
25. Describe both the structural and functional differences between the optic disc and macula. (p. 508)
26. State the three chambers of the eye and their contents. (p. 511)
27. Describe the production and flo of the aqueous humor. (p. 511)
28. Explain the principles of the corneal and papillary reflex. (p. 508)
29. State the differences between the external and internal corneal epithelium. (p. 509)
30. Describe the structure and function of the ciliary body. (p. 509)
31. State the two muscles forming the iris. (p. 509)
32. State the 4 types of retinal neurons. (p. 510)
33. State the average value of intra-ocular pressure. (p. 510)
34. State the optic power of the lens and cornea. (p. 509, 510)
35. State the 3 arteries supplying the macula. (p. 511)

VI. Accessory visual structures

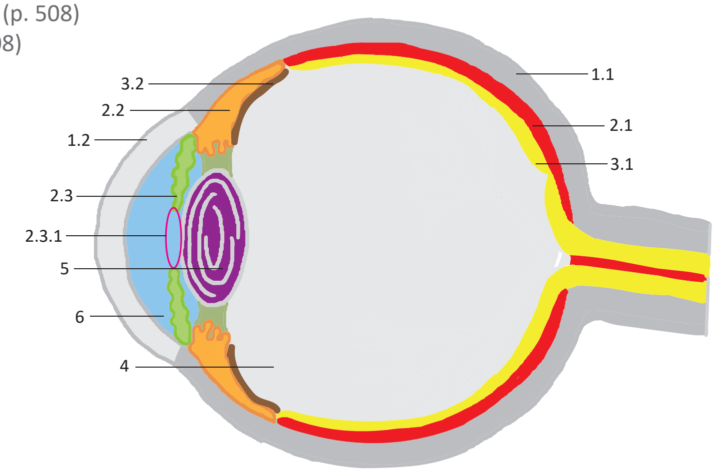
36. State the 6 accessory visual structures. (p. 512)
37. State the three types of glands located in the eyelids. (p. 512)
38. State the 5 parts of the eyeball covered with the conjunctiva. (p. 512)
39. Draw a scheme of the eyeball movements of the individual extra-ocular muscles. (p. 513)
40. Explain the terms divergence and convergence (of the eye). (p. 513)
41. State the 6 extra-ocular muscles and specify their innervation. (p. 513)
42. Describe the production, circulation and drainage of tears. (p. 514)

VII. Integumentum (Skin)

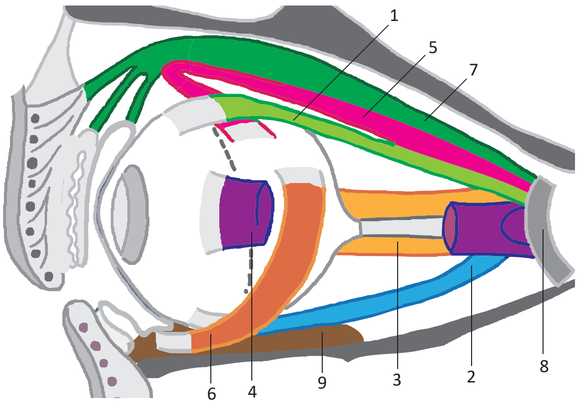
43. State the 7 functions of the skin. (p. 515)
44. Explain the clinical significance of the skin tension lines. (p. 515)
45. State and describe the three types of the skin glands. (p. 515)
46. Classify the types of hair and specify their location. (p. 516)
47. Explain how and from where the nail grows. (p. 516)

VIII. Breast and mammary gland

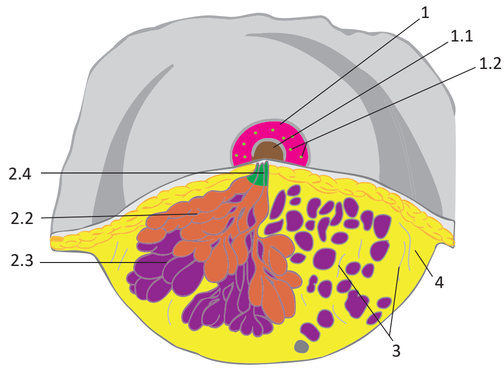
48. State which type of gland the mammary gland is. (p. 517)
49. State the 4 types of adult breast. (p. 517)
50. State the lymph node groups draining the breast. (p. 517)



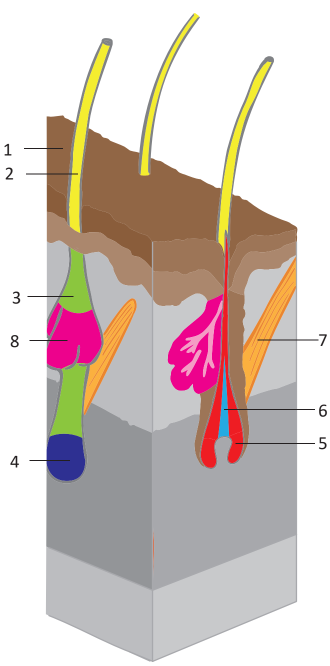
Describe the structure of the eyeball



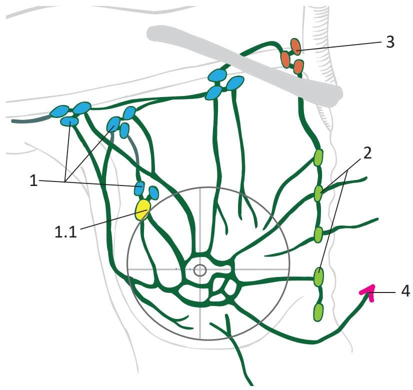
Describe the eye muscles



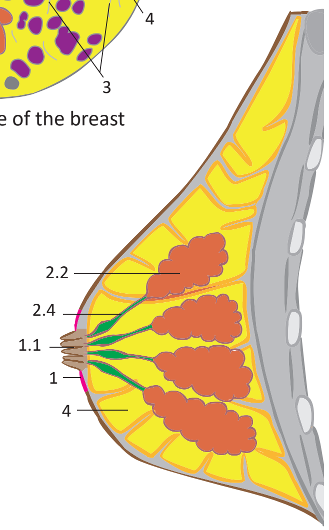
Describe the structure of the breast



Describe the composition of the skin and its derivates



Describe the lymphatic drainage of the breast



Describe the structure of the breast

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Anatomists

Assoc. prof. Nihal Apaydın, MD – Department of Anatomy, Faculty of Medicine, Ankara University, Ankara, Turkey
Assoc. prof. Václav Báča, MD, PhD – Department of Anatomy, Third Faculty of Medicine, Prague, Czech Republic
Assoc. prof. Květuše Lovásová, VDM, PhD – Department of Anatomy, Pavol Jozef Šafárik University, Košice, Slovakia
Pavel Šnajdr, MD, PhD – Department of Anatomy, First Faculty of Medicine, Charles University, Prague, Czech Republic

Clinicians

Dale Kalina, MD – Department of Internal Medicine, University of Saskatchewan, Canada

Medical students

Linda Kašičková
Adéla Kuklová

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